LMNOP AVIATION FINAL REPORT

Practitioners: Sarah Chapin, Ana Jackson, Michelle Sinn, and Patrick Smith
OPWL 529: Needs Assessment
Boise State University
Fall 2025

Table of Contents

Academic and Confidentiality Disclaimer

This report has been anonymized and generalized for academic and portfolio purposes. All company names, proprietary tools, internal metrics, and identifying details have been removed or replaced with pseudonyms to protect organizational confidentiality. While the analysis methods and recommendations reflect authentic practices and aim to mirror real-world project work, certain organizational details have been modified or fictionalized to maintain privacy.
Because confidentiality was prioritized, some granular details may be omitted or generalized. Readers should understand that this report represents a rigorous academic approach to applied business analysis, but not all aspects reflect an actual company or its operations.

Executive Summary

Report: LMNOP Aviation Needs Assessment Business Unit: Domestic Service Center Network
Assessment Window: 2 months (remote data collection)

Prepared by: Team Aviation, Boise State University OPWL Masters Program

Confidentiality Statement: This report provides a generalized academic analysis based on real business practices, with details modified to ensure confidentiality.

Purpose and Context

Leadership requested a needs assessment to address productivity and customer service challenges. These issues impact the Service Division's strategic goals: improving profitability, efficiency, customer experience, and capacity.
The assessment focused on two key performance areas with significant gaps between current results and targets.

- **Productivity**: Less than half of sites are meeting their sites productivity goals
- **Invoice Accuracy**: Half of the sites are meeting invoice accuracy goals
- **Target**: 90% of sites meeting both of those goals

Why It Matters

These gaps are consistent with measurable impacts in:

- Non-billable labor and rework
- Revenue leakage from invoice errors
- Delays that constrain capacity and planning
- Customer churn risk driven by billing friction and reduced trust

What We Assessed

We reviewed performance across sites and engaged stakeholders across the network, including the VP of Domestic Service Center Network (executive sponsor), GMs, Maintenance Managers, TSMs, CSMs, Technicians, and the Talent & Development Manager (project sponsor).

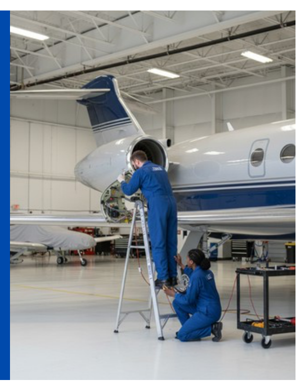
How We Did It

- Frameworks: We applied Chevalier's Behavioral Engineering Model and Rummler & Brache's Nine Variables framework to guide our needs assessment, shape our questions, and identify targeted interventions.
- Interviews: **Eight interviews** with TSMs and CSMs at exemplary and non-exemplary sites
- Surveys: **300 surveys** collected across GMs, MMs, TSMs, CSMs, and technicians at all sites
- Light **systems review**: Computer tools and systems used daily
- **Data review**: Productivity, invoice accuracy, headcount, attrition, years of experience, job descriptions

Constraints: Remote-only collection (no onsite observations), sampling due to time, potential response bias, limited review of support departments, no competitive benchmarking, no access to compensation data.

Key Finding

Differences in performance across sites reflect less about individual effort and more about whether sites are consistently enabled with standard work, usable systems, mentoring, and sufficient role capacity. High-performing sites have developed practical routines that strengthen planning, manage variability, and reduce rework; other sites have not yet built the same consistency.



What We Discovered (Root Cause Themes)

Across data sources, seven recurring drivers explain most of the productivity and invoice accuracy challenges:

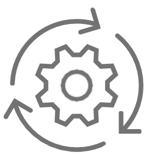
- **Inconsistent standard work processes (SOP gaps)**: Core maintenance and invoicing processes vary site-to-site (scoping, capacity planning, communication, handoffs, tool tracking, invoice review), leading to inconsistent cycle times, inefficient utilization, and preventable invoice errors.
- **Fragmented information**: Technical data and project communication live across multiple systems and informal channels, requiring time-consuming searches and increasing missed updates, rework, and invoice inconsistencies.
- **Tools and digital workflows add friction**: "Paper-free" systems are experienced as clunky and manual; validation is limited; after-hours support is constrained; training and change communication are inconsistent; configuration differences create uneven visibility into hours and job status.
- **Uneven capability building and knowledge transfer**: Technicians and CSMs largely learn through informal methods; competency expectations and onboarding are not standardized, making capacity planning difficult; off-shift patterns reduce coaching; cross-training is limited, increasing handoffs and rework.
- **Variable performance management and misaligned incentives**: Accountability and feedback practices vary by leader/site; underperformance is not consistently addressed; pay compensation in leadership roles undermines advancement incentives and weakens sustained performance culture.
- **Physical environment and equipment**: Some sites are impacted by outdated tools, equipment condition/calibration, facility constraints, and unreliable internet increasing waiting, searching, and improvising, reducing productive maintenance time.
- **Role design and capacity constraints**: Invoicing responsibilities are often layered onto already full coordination/leadership roles; "multiple hats" reduces time available for thorough, consistent invoice review.

Helpful Hint

Click on each title for links to detailed evidence in the report.

Recommendations

Using **multi-criteria analysis and SWOT,** the team recommends three integrated improvements:



Standardize processes (SOPs / standard work) Network-wide standards for planning/scheduling, project management, work allocation and handoffs, invoice review/QC, and cross-functional communication with a clear system of record.

Improve tools and systems
Centralize/normalize documentation access; assess system usability/configuration/validation; reduce high-friction tasks; address off-shift support constraints.

Formalize knowledge transfer
Mentoring, role-specific onboarding, a competency model with certification pathway for technicians, and targeted cross-training (especially across TSM/CSM workflow dependencies).

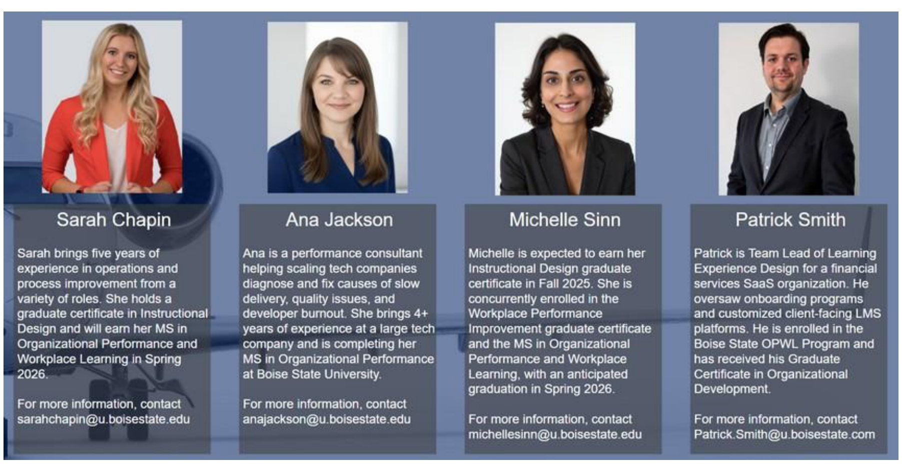
Timeline

We will implement interventions through a phased 12–18 month plan that balances ambition with practicality and delivers early wins. The focus is on three primary interventions mentioned above:

- **Standardize processes**
- **Improve tools and systems**
- **Formalize knowledge transfer across sites**

- **Phase 0:** Build the implementation team.
- **Phase 1:** Standardize SOPs as the foundation for all other work.
- **Phase 2:** Upgrade tools and systems for easy access to SOPs and key information.
- **Phase 3:** Formalize knowledge transfer processes.
- **Phase 4:** Implement second-level recommendations (performance management, career development).
- **Phase 5:** Third-level recommendations will be applied selectively where relevant.

Team Aviation



Purpose and Scope

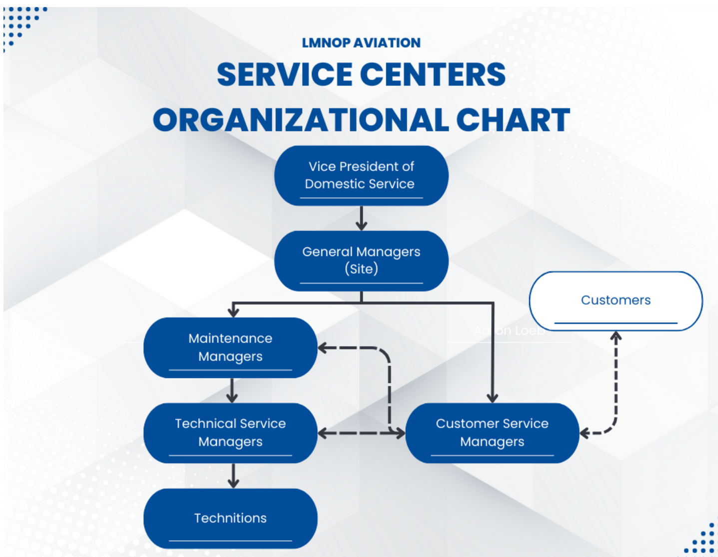
The purpose of this needs assessment is to identify systemic factors affecting productivity and invoice accuracy across the service network and to provide actionable recommendations for improvement. Using triangulated data sources and structured frameworks such as Chevalier's Behavior Engineering Model (BEM) and Rummler & Brache's Nine-Variable Model, this assessment seeks to uncover root causes of performance gaps and prioritize interventions that address critical areas of information, resources, and knowledge. This report focuses on organizational, process, and individual-level factors influencing operational performance within the service center network. The scope includes:

- Analysis of current practices, workflows, and performance expectations.
- Identification of gaps in leadership, training, and process standardization.
- Evaluation of environmental and systemic constraints impacting productivity and invoice accuracy.
- Development of a phased implementation roadmap with recommendations.

Although there are some limitations, such as remote data collection and lack of competitive benchmarks, these do not affect the breadth or depth of this report. The findings and recommendations are grounded in validated frameworks, sound academic principles, and data-driven analysis, ensuring credibility and actionable results.

Organization

Figure 1
LMNOP Aviation Service Center Organizational Chart



LMNOP Aviation is a global provider of aircraft maintenance and support, operating a network of specialized service facilities. The company focuses on keeping aircraft safe, reliable, and ready to perform through practical, high-quality solutions. Guided by its core values of precision, partnership, and progress, LMNOP aims to make every customer interaction smooth and stress-free.

The service division offers a wide range of capabilities, including predictive maintenance programs, rapid-response repair services, performance upgrades, lifecycle support planning, and digital tools that simplify workflows. These services are designed to reduce downtime, improve efficiency, and help customers get the most out of their aircraft. Each service center acts as both a revenue hub and a key customer touchpoint, ensuring compliance and operational readiness while delivering an experience that builds trust.

Strategic objectives for the service division include improving customer satisfaction through consistent, high-quality service; streamlining operations to enhance efficiency and reduce costs; developing workforce capability through structured training and knowledge transfer; and strengthening profitability across the network. These objectives align with broader organizational priorities focused on operational excellence, innovation, and long-term customer value. Progress toward these goals is measured through key performance indicators such as productivity and invoice accuracy, ensuring accountability and continuous improvement.

The aviation maintenance sector shares operational parallels with other service industries, balancing rapid-turnaround tasks with complex, long-duration projects. Service centers typically include senior operations leaders and technical managers overseeing customer service and maintenance activities. Each location generally comprises a General Manager, Maintenance Manager, Customer Service Managers, Technical Service Managers, and Technicians.

Roles

Vice President of Domestic Service Center Network

The Vice President of Domestic Service provides strategic direction and operational oversight of domestic service center locations This role is responsible for network-wide performance across key metrics, including productivity, invoice accuracy, customer satisfaction, and profitability. The VP collaborates with General Managers at each location,to set organizational goals and standards, and drive initiatives to improve operational excellence and capacity across the service center network.

General Manager (GM)

General Managers lead overall service center operations including strategic planning, resource allocation, financial performance, and coordination of service center roles. They ensure their centers meet organizational targets for productivity, invoice accuracy, customer satisfaction, and profitability, while managing relationships with corporate leadership and customers.

Maintenance Manager (MM)

Maintenance Managers support the GM by overseeing day-to-day maintenance operations, coordinating workflow across technical teams, and ensuring operational targets are achieved. They work closely with Technical Service Managers to address scheduling challenges, parts delays, and project complications that affect productivity and throughput.

Customer Service Manager (CSM)

Customer Service Managers handle customer communication, relationship management, invoicing, dispute resolution, and credit processing. They collaborate with MMs and TSMs to ensure accurate billing based on completed work, manage customer expectations throughout the maintenance process, and resolve invoice discrepancies. Their work directly influences invoice accuracy rates and customer satisfaction.

Technical Service Manager (TSM)

Technical Service Managers oversee teams of technicians to execute aircraft maintenance projects from start to finish. TSMs are responsible for parts ordering, project planning and execution, troubleshooting, problem-solving, work allocation, and ensuring timely completion. Their performance impacts productivity and overall service quality.

Technician (Tech)

Technicians perform aircraft maintenance, repair, and inspection work on aircraft. They execute work orders, document labor hours, return unused parts, troubleshoot technical issues, and ensure quality standards are met. Technician productivity is a key driver of service center performance.

Project Sponsors and Key Stakeholders

This initiative is sponsored by Blake Harrison, Talent and Development Manager. Blake provides strategic guidance and ongoing support throughout the project lifecycle. He reviews and offers feedback on data, reports, and findings, and will ultimately be responsible for advancing the project into its implementation phase once solutions are finalized and presented.

The project was initiated by Cameron Steele, Vice President of the Domestic Service Center Network. He believes that enhanced training for Technical Service Managers and CUstomer Service Managers, as well as improved project planning, are key to driving meaningful change. His feedback on previous improvement efforts will help shape data-driven solutions that align with operational goals.

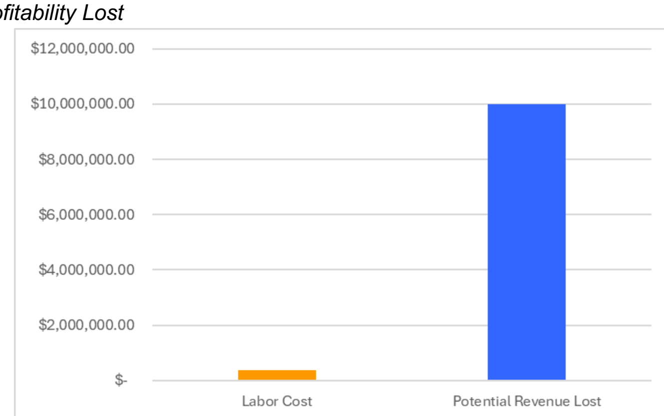
Additional key stakeholders include General Managers across the network. These leaders play a critical role in encouraging honest employee feedback and demonstrating visible support for the initiative. Their voices will help promote engagement and validate the project’s importance.

The Problem

Team Aviation was asked to address productivity and customer service challenges within LMNOP Aviation. Upon reviewing LMNOP Aviation’s metrics, a significant performance gap was identified in two key areas: Productivity and Invoice Accuracy. These metrics are tied to the service division’s strategic goals of improving profitability, efficiency, customer experience, and capacity across service centers. Currently, only a minority of centers meet productivity goals, and less than half are meeting designated invoice accuracy thresholds.

1.Profitability

Figure 2
Potential Profitability Lost



Low productivity reduces profit margins. When technicians require more hours than planned to complete maintenance work, labor costs exceed estimates, reducing profitability per project. Invoice errors compound this problem, as inaccurate billing leads to revenue loss through undercharging or costly disputes and write-offs when overcharging occurs. Across underperforming centers that handle hundreds of maintenance events annually, these inefficiencies represent substantial lost revenue and increased operational costs. The calculation below displays a potential estimated annual impact of invoice accuracy and how it could affect an organization:

Assumptions:

- Avg invoices per year: 13,500
- Avg count of credits per year: 1,000
- Invoice error rate: 7.4%
- Potential hourly rate for CSM: $60
- Potential time to resolve each credit: 2-6 hours
- Potential average cost of each credit: $1,500 - $10,000

Labor cost: *$120,000 - $360,0000 (Potential CSM time spent resolving credits at $60/hour for 2-6 hours per issue)*

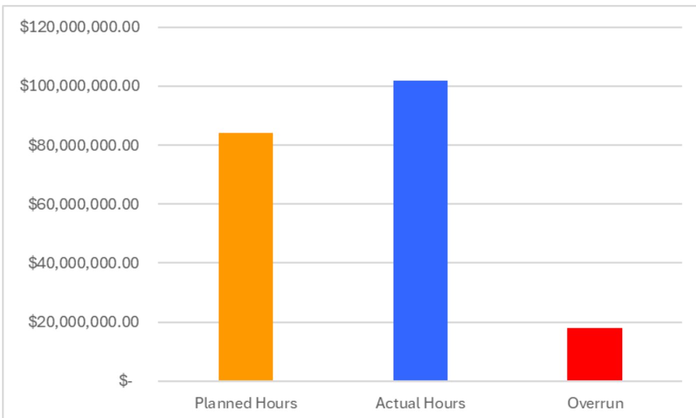
Revenue loss: *$1,500,000 - $10,000,000 (Assuming credits could vary between $1,500-$10,000)*

***Total potential impact:*[1]** *$1,620,000 - $10,360,000 annually*

2.Operational Efficiency

Figure 3

Operational Efficiency Costs



Inconsistent productivity creates unpredictable project timelines, making capacity planning and resource allocation difficult. When service centers underperform, the organization cannot reliably forecast how long maintenance will take or how many aircraft it can service simultaneously. Invoice errors require rework, as CSMs must correct invoices, communicate with customers about discrepancies, and reprocess billing. This diverts time delaying projects and affecting scheduling for subsequent customers, creating bottlenecks across the network. The calculation below displays a potential estimated annual impact of productivity and how it could affect an organization:

Assumptions

- Total Planned Hours: 1,400,000
- Total Actual Hours: 1,700,000
- Variance: 17%
- Average cost per labor hour: $60

Labor exceedance calculation: *1,700,000 - 1,400,000 = 300,000*

***Estimated cost of overrun:*[2]** 300,000 x $60 = $18,000,000 annually

[1] These figures are directional and based on broad assumptions; actuals may differ significantly.

[2] These figures are directional and based on broad assumptions; actuals may differ significantly.

3.Customer Experience

Customer satisfaction is directly affected by both metrics. Extended maintenance times delay aircraft return to service, disrupting customer operations and travel plans. Invoice errors damage trust as customers question competence when billed incorrectly, and disputes over charges can create friction in relationships. In a competitive aviation maintenance market where customers have alternatives, poor performance on these dimensions pushes clients toward competitors, threatening LMNOP's market position and reputation.

4.Capacity and Growth

Underperformance in productivity limits the organization's ability to serve additional customers or expand operations. When centers operate below productivity targets, they cannot handle the volume of maintenance work that the organization's capacity should theoretically support. This constrains revenue growth and prevents LMNOP from capitalizing on market opportunities. Additionally, the time spent correcting invoice errors further reduces effective capacity, as CSMs dedicate hours to reviewing billing rather than processing new business.

Strategic Impact of Closing Operational Gaps

Resolving productivity and invoice accuracy gaps will enable LMNOP Aviation to achieve strategic objectives and strengthen its competitive advantage. Benefits include:

- Enhanced Profitability: Increased revenue through accurate billing, reduced labor costs, and improved margins.
- Operational Excellence and Predictability: Reliable capacity planning, optimized resource allocation, and faster turnaround times.
- Superior Customer Experience and Retention: Accurate billing builds trust, faster service minimizes disruption, and consistent quality drives loyalty.
- Increased Capacity for Growth: Higher throughput enables servicing more aircraft without facility expansion, unlocking revenue growth.
- Organizational Capability and Knowledge Retention: Structured onboarding, documented processes, and continuous improvement culture strengthen institutional resilience.
- Employee Engagement and Retention: Clear career pathways, better tools, and consistent feedback improve motivation and reduce turnover.
- Competitive Market Position: Operational excellence differentiates LMNOP, attracting customers and talent while enabling pricing flexibility and scalability.

Data Collection Methods

Team Aviation employed a multi-faceted approach to ensure comprehensive coverage of potential performance barriers.

- **Interviews:** Conducted with TSMs and CSMs at exemplary and non-exemplary sites.

- **Surveys:** Distributed to all GMs, MMs, TSMs, CSMs, and technicians across the organization.
- **Extant Data:** Collected various extant data for all locations.

Frameworks and Models

Team Aviation applied two complementary frameworks to properly identify performance barriers and ensure proper interventions were applied.

Chevalier's Variant of the Behavior Engineering Model (Chevalier's BEM)

Team Aviation utilized the Chevalier's Behavior Engineering Model, a variation of Thomas Gilbert's original BEM model, developed by Roger Chevalier (2003), to systematically identify performance barriers. The BEM categorizes factors as environmental supports (factors in the work environment that enable or impede performance) and individual knowledge, skills, and abilities (what the performer brings to the situation).

The model examines six variables in a priority order:

Environmental Factors (which typically account for 75% of performance problems):

- Information: Clear expectations, relevant feedback, performance guides, and performance management systems.
- Resources: Materials, tools, time, documented processes, and a conducive work environment.
- Incentives: Financial and non-financial rewards, job enrichment, career development opportunities, and positive work culture.

Individual Factors:

- Knowledge/Skills: Necessary competencies, proper placement, and cross-training
- Motives: Alignment between employee values and work requirements, desire to perform
- Capacity: Ability to learn and perform, proper selection and placement, freedom from emotional limitations

This prioritization reflects research showing that environmental factors are more cost-effective to address than individual factors, and that fixing individual factors without addressing environmental barriers rarely produces sustainable improvement. Figure 4 visually displays this framework and factors. In our data analysis section we dive into the colors and the questions within this graphic.

Figure 4

Chevalier's BEM - Question Guidebook

Environmental Factors	Information	Resources	Incentives
	- Do employees clearly understand their roles and responsibilities?	- Do employees have access to the tools and matials needed to perform their jobs?	- What type of rewards or recognition are currently in place?
	-Are performance expectations communicated effectively?	- Are there resource shortages that impact performance?	- Do employees feel their efforts are acknowledged and valued?
	- How frequently is feedback provided?	- Is the work environment organized and conducive to productivity?	- Are there opportunities for career growth or job enrichment?
	- Are there guidelines and or documentation available to support task execution?	- Are processes clearly defined and accessible?	- How does the organization support employee motivation?
Individual Factors	Knowledge/Skills	Capacity	Motives
	- Do employees have the necessary training and experience for their roles?	- Do employees feel physically and metally capable of performing their duties?	- Do employees feel personally motivated to perform well?
	- Are there gaps in knowledge or skills that affect performance	- Are job demands reasonable for the employees abilities	- Are individual goals aligned with oragnizational goals?
	- Is ongoing training avaiable and accessible?	- Are accommodations available for employees with specific needs?	- What aspects of the job do employees find most fulfilling or frustrating?
			- Were employees recruited witha clear understanding of the job expectations?

Rummler and Brache Nine-Variable Framework (Rummler and Brache's Nine-Variable Model)

Influenced by Gilbert's Behavior Engineering Model, Rummler and Brache proposed Nine-Variables to analyze organizational performance across three levels: Organizational, Process, and Performer (Stefaniak, 2010). Their framework examines how performance occurs through three lenses: Goals, Design, and Management.

- Goals: Alignment between organizational vision, process goals, and job-level goals.
- Design: Resource allocation and structure at organizational, process, and job levels.
- Management: Oversight and feedback mechanics ensuring alignment and accountability.

This framework enabled Team Aviation to examine how organizational decisions cascade through processes to create job-level barriers, revealing that performance issues at individual service centers often stemmed from organizational and process-level factors rather than performer-level problems alone. Figure 5 is a visual representation of Rummler and Braches Nine-Variable Framework. As with Figure 4, the colors and questions will be explained in the data analysis section.

Figure 5

Rummler and Brache's Nine-Variable Model - Question Guidebook

Top Row: Organizational Level	Organization Goals	Organization Design	Organization Management
	- What are the strategix objectives of the organization?	- Is the organizational structure coducive to achieving strategic goals?	- How does leadership monitor organizational performance?
	- How are these goals communicated across departments?	- Are departements and teams aligned and coordinated effectively?	- What systems are in place for decision-making and governance?
	- Are performance metrics aligned with strategic goals?	- Are there clear lines of authority and accountability?	- How are resources allocated across the organization?
	- How does leadership prioritize these goals?	- How flexible is the organization in adapting to change?	How does leadership respond to performance issues?
Middle Row: Process Level	Process Goals	Process Design	Process Management
	- What are the expected outcomes of this process?	- Is the process clearly defined and documented?	- Who is responsible for managing this process?
	- How are process goals communicated to employees?	- Are roles and responsibilities within the process well understood?	How is performance monitored and reported?
	- Are there metrics in place to measure process success?	- Are there bottlenecks or inefficiencies in the workflow?	- Are there feedback loops to improce the process?
	- How are the process goals aligned with organization goals?	How often is the process reviewed or updated?	- What happens when the process fails or underperforms?

Data Analysis Methods

Extant Data Review

Team Aviation used anonymized metrics regarding the organization's sites. The primary data analyzed was Productivity and Invoice Accuracy. Secondary metrics reviewed included Schedule Adherence, Customer Satisfaction, Hours to Planned, Billing and Labor Efficiencies, NOP, and NOP divided by headcount.

Codebook Development and Analysis

Team Aviation developed a structured codebook using Chevalier's Behavior Engineering Model (BEM) and Rummler & Brache's Nine-Variable Model, this allowed a structured way to analyze qualitative data from TSM and CSM interviews and organization-wide surveys.

- **Productivity-related codes** aligned with BEM's six variables.
- **Invoice Accuracy-related codes** aligned with Rummler & Brache's Nine-Variable Model.

This theory-driven approach ensured consistent classification across data sources and revealed systemic patterns rather than isolated issues. Multiple codes were applied when issues spanned categories.

Cause Analysis

Root Causes and Five Whys

We used a structured, evidence-based root cause process to ensure we focused on the drivers most likely to improve both productivity and invoice accuracy.

The resulting "performance differentiators” served as the starting point for five whys.

1. ***Identify Performance Differentiators***

   We began by comparing exemplary and non-exemplary sites to identify factors associated with high and low performance (Table 1). These differences were triangulated across:

   - Interviews and Surveys (coded by BEM and Nine-Variable Model frameworks)
   - Extant operational data

Table 1
Factors Influencing Project Needs

Need	Factors Contributing to Desired State of Performance	Factors Inhibiting Desired State of Performance
1. Improve productivity	● Having more experienced employees ● Having tech crews cross-trained on all aircraft models ● Effective and proactive planning for maintenance events ● Effectively managing workflow when an unexpected maintenance event happens ● Lead technicians proactively communicating with TSMs ● One central handoff template that is consistently updated and all stakeholders have access to ● Strong mentorship/learning culture ● Culture of efficiency and taking pride in the work ● Keeping equipment well-maintained ● Actively monitoring technician hours/performance ● TSMs providing frequent feedback on technician performance ● TSMs spending at least half of their time out on the floor interacting with leads and technicians (not just at their desks) ● More experienced employees being available to answer questions across shifts ● Senior management being responsive to needs ● Taking steps to decrease or manage high workload in a sustainable way for employees	● A growing percentage of new and less experienced technicians ● Manuals/Information spread across multiple portals ● Tools or pieces/parts of a customer's plane being misplaced ● Ineffective planning for maintenance events ● Inconsistent or inaccurate communication between cross-functional teams and/or shifts ● Communication/Information about one project is spread across multiple tools with inconsistent access across stakeholders ● Some systems are not user-friendly and requires frequent updates and time-consuming, manual processes to use ● Varying proficiency with new computer-based tools ● Lack of standardized onboarding ● Lack of accountability/monitoring technician hours ● The clocking program requires technicians to be clocked on a job, which some misuse to hide hours ● Design, lack of, and/or inefficiently managing hangar space ● Slow internet or outages at the facilities ● Unavailability of parts (due to shortages, tariffs, shipping times) ● Employees at centers with lower head count perform multiple roles, leading to very high workloads and task saturation

2. Improve invoice accuracy	• CSMs with a technical background in aviation maintenance • Proactive and accurate communication with the customer • Strong communication and collaboration between TSMs and CSMs • TSMs consistently ensuring parts are returned and hours are accurately logged	• CSMs without a technical background in aviation maintenance • High workload • No job aid or reference for technical codes (must be memorized) • System is slow to reflect whether a customer has a warranty or other maintenance coverage terms • Communication/Information about one project is spread across multiple tools with inconsistent access across stakeholders • Varying proficiency with computer-based tools • Job responsibilities vary by site/dependent on headcount, increasing employee workload

2. ***Apply Five Whys Analysis***

To move beyond surface systems, we applied a Five Whys approach to isolate underlying conditions driving these differentiators. This ensured that identified causes were systemic.Separate chains for productivity and invoice accuracy identified root causes within the project's scope. Rather than treating each cause as isolated, we clustered similar items into broader themes, including:

- Digital tools and system design
- Information and communication management
- Process and SOP standardization
- Training and capability systems
- Performance management and culture

These themes align with Chevalier's Updated Behavior Engineering Model (Environment: Information, Resources, Incentives; Individual: Knowledge/Skills, Capacity, Motives) and highlight where systemic issues are concentrated. Grouping causes into themes allowed us to:

- Target overlapping issues impacting both productivity and invoice accuracy

- Distinguish cross-site themes from site-specific causes
- Create a clear bridge to intervention selection, feeding directly into SWOT and multi-criteria analyses

3. ***Group and Map Causes***

   Related causes were grouped into broader themes to synthesize patterns across sites and roles. Many themes were interdependent so to ensure we were considering the full picture, we analyzed them using a matrix that mapped each cause to the two target outcomes: Productivity and Invoice Accuracy (Table 2 and 3)

   - These visuals make cross-cutting drivers visible (Table 4).
   - Prioritized highest ROI causes that would improve both metrics

Table 2
Five Whys Summary (Productivity)

Problem Statement: Less than half of the centers are meeting their individual productivity goal set by the organization.			
Root Cause	**Chevalier's BEM Focus**	**Nine-Variable Model Focus**	**Theme**
Processes and maintenance project management (organizational SOPs) are not standardized across sites, so managers decide on their own how to proceed in most situations. This has led to idiosyncratic processes that don't always work as intended.	• Environment - Resources • Environment - Information	• Organization Design • Process Design	Process and SOP Standardization
Manuals and job information are spread across multiple portals with no simple, standardized access path. This leads to	• Environment - Resources • Environment - Information	• Process Design • Performer Design	Information and communication management

inconsistent processes, time to completion, and costs.			
Training is inconsistent across centers, and there is no formal knowledge transfer process. This leads to slower and uneven ramp-up for new and less-experienced technicians.	• Environment - Resources (training & mentoring systems) • Individual - Knowledge/Skills	• Organization Design • Process Design	Training, onboarding, and capability systems for key roles
There is no consistent, organization-wide way to document and maintain technician competencies and task capabilities (e.g., skills matrix or grade-level standards). This leads to variability in assignment quality and efficiency by site and manager.	• Environment - Information • Environment - Resources	• Organization Design • Process Management	Training, onboarding, and capability systems for key roles
Computer tools are not user-friendly, require time-consuming manual steps, and undergo frequent, poorly communicated updates. This leads to proficiency gaps and inefficiencies across sites.	• Environment - Resources • Environment - Information • Individual - Knowledge/Skills	• Process Design • Performer Design	Digital Tools and System Design
There are not enough of the right tools at point of use; tools are not consistently maintained or replaced	• Environment - Resources • Environment - Information	• Process Design • Organization Management	Physical environment and equipment

Hangar design constraints and limited space require reshuffling aircraft or redirecting incoming aircraft when the hangar reaches capacity. This leads to release delays and capacity issues.	• Environment - Resources	• Job/Performer Environment • Organization Design	Physical environment and equipment
Manager feedback quality varies significantly by individual manager; some provide specific, timely coaching while others give vague or infrequent feedback. This leads to limited technician development and reduced productivity	• Environment - Information • Individual - Knowledge/Skills (coaching skills)	• Performer Management • Performer Design • Organization Management	Performance Management and Culture

Table 3
Five Whys Summary (Invoice Accuracy)

Problem Statement: Half of the sites are not meeting their invoice accuracy goal set by the organization.			
Root Cause	**Chevalier's BEM Focus**	**Nine-Variable Model Focus**	**Theme**
No standard processes for invoice review and quality control; sites and individuals handle review differently, and some invoices receive little or no systematic check before being issued. This leads to inconsistent invoice accuracy and higher error rates.	• Environment - Resources • Environment - Information	• Process Design • Process Management	Process and SOP Standardization

Lack of documentation and practical guidelines for invoice review, so CSMs lack clear written criteria or checklists and rely on personal judgment and habits. This leads to variability in review quality and missed errors.	• Environment - Information • Environment - Resources	• Process Design • Performer Design	Process and SOP Standardization
No standardized onboarding or training curriculum for CSM or invoice-related roles. This leads to inconsistent knowledge and uneven performance across sites.	• Individual - Knowledge/Skills • Environment - Resources	• Performer Design • Process Management	Training, onboarding, and capability systems for key roles
At some sites, TSMs also perform CSM and invoicing responsibilities, leaving insufficient time for thorough invoice review. This leads to rushed checks and increased likelihood of mistakes.	• Individual - Capacity • Environment - Resources (lack of time; pressure to keep invoices moving)	• Performer Design • Organization Design	Role Design and capacity
CSMs are hired and developed primarily for customer service, coordination, and administrative skills, and receive limited structured training on aircraft systems, maintenance concepts, and how those map to invoice line items. This leads to gaps in technical understanding and inaccurate billing.	• Individual - Knowledge/Skills • Environment - Resources	• Performer Design	Training, onboarding, and capability systems for key roles
Computer tools are not user-friendly; confusing or missing fields and clunky	• Environment - Resources	• Process Design	Digital Tools and System Design

navigation make it easy to miss or mis-enter information under time pressure. This leads to data entry errors and incomplete invoices.	• Environment - Information	• Performer Design	
Weak accountability culture around invoice accuracy: expectations are not consistently reinforced, and there is little follow-up, coaching, or consequences when error rates are high. This leads to persistent quality issues and lack of improvement.	• Environment - Information • Individual - Motives	• Performer Management • Organization Management	Performance Management and Culture
System design of coding and billing tools allows errors to pass through uncaught; many incorrect or incomplete combinations are allowed, and invoices can be finalized without robust validation. This leads to systemic inaccuracies and customer dissatisfaction.	• Environment - Resources • Environment - Information	• Process Design	Digital Tools and System Design
No standardized cross-functional communication workflow or designated system of record for maintenance project information; teams use multiple channels (email, text, phone, ad hoc notes) and rely on individuals to piece together what was done for invoicing. This leads to incomplete or incorrect invoice details and delays.	• Environment - Information • Environment - Resources	• Process Design • Process Management	Information and communication management

Table 4

Cause-Problem Overlap Analysis

Theme	BEM Category	Productivity Root Cause	Invoice Accuracy Root Cause	Shared Across Both Problems?	Effects Across All Sites
Process and SOP Standardization	Information; Resources	Processes and maintenance project management (organizational SOPs) are not standardized across sites, so managers decide on their own how to proceed in most situations. This leads to inconsistent processes and unpredictable outcomes.	Weak or no processes for invoice review and quality control; sites and individuals handle review differently, and some invoices receive little or no systematic check before being issued. This leads to inconsistent invoice accuracy and higher error rates. Lack of documentation and practical guidelines for invoice review, so CSMs lack clear written criteria or checklists and rely on personal judgment and habits. This leads to variability in review quality and missed errors.	Yes – both causes show “each site does it differently, with no clear standard”	Yes
Information and communication management	Information; Resources	Manuals and job information are spread across multiple portals with no simple, standardized access path. This leads to	No standardized cross-functional communication workflow or designated system of record for maintenance project information; teams use multiple channels (email, text,	Yes - same pattern of fragmented information, different points in the workflow	Yes

		inconsistent processes, longer completion times, and higher costs.	phone, ad hoc notes) and rely on individuals to piece together what was done for invoicing. This leads to incomplete or incorrect invoice details and delays.		
Digital Tools and System Design	Information; Resources; Knowledge/ Skills	Computer tools are not user-friendly, require time-consuming manual steps, and undergo frequent, poorly communicated updates, and proficiency with these tools varies across sites. This leads to inefficiencies and increased risk of errors.	Computer tools are not user-friendly; confusing fields and codes and clunky navigation make it easy to miss or mis-enter information under time pressure. This leads to data entry errors and incomplete invoices. System design of coding and billing tools allows errors to pass through uncaught; many incorrect or incomplete combinations are allowed, and invoices can be finalized without robust validation. This leads to systemic inaccuracies and customer dissatisfaction.	Yes - same digital/tool family slows work and lets errors through	Yes
Training, onboarding, and capability systems for key roles	Knowledge/ Skills; Resources	Inconsistent training across centers and a lack of formal knowledge transfer systems mean new and less-	No standardized onboarding or training curriculum for CSM or invoice-related roles. CSMs are hired and developed primarily for customer service,	Yes - same structural issue: complex roles, informal learning, and weak capability systems	Yes

		experienced technicians ramp up more slowly and inconsistently. This leads to uneven productivity and higher error rates. There is no consistent, organization-wide way to document and maintain technician competencies and task capabilities. This leads to variability in assignment quality and efficiency by site and manager.	coordination, and administrative skills, and receive limited structured training on aircraft systems, maintenance concepts, and how those map to invoice line items. This leads to gaps in technical understanding and inaccurate billing.		
Performance management and culture	Information; Resources; Knowledge/ Skills	Manager feedback quality varies significantly by individual manager; some provide specific, timely coaching while others give vague or infrequent feedback. This leads to limited technician development and reduced productivity.	Weak accountability culture around invoice accuracy: expectations are not consistently reinforced, and there is little follow-up, coaching, or consequences when error rates are high. This leads to persistent quality issues and lack of improvement.	Yes - in both areas, errors or performance issues are not consistently coached for followed-up on	No (Site and manager-dependent)

Productivity Cause Themes

Theme 1: Physical environment & equipment

Across several sites, productivity is constrained by gaps in the physical work environment rather than by technician effort or skill. Technicians often lack the right tools at point of use, encounter equipment that is out of calibration or in poor condition, and work in facilities that are too small or poorly organized. Basic resources such as diagnostic laptops, specialty tools, and reliable Wi-Fi are not consistently available, leading some technicians to buy their own tools or spend time searching across cribs, hangars, or outdoor storage.

These conditions create frequent slowdowns, waiting on parts, reshuffling aircraft in tight hangars or tracking down shared tools.Time is also lost to troubleshooting unreliable diagnostic equipment, and coping with slow or unstable internet access. As a result, a significant portion of the workday is consumed by non-value-added activities such as waiting, searching, and improvising around facility and equipment limitations, reducing productivity.

Evidence

Survey results

- **Tool condition and availability:** 40.4% of technicians agreed that equipment and tools are maintained and in good working condition, and 55.6% indicated they often or always have the right tools for the job in good condition. At the same time, 39.3% of technicians reported having to search for the right tool or equipment multiple times a week or every day.
- **Parts delays:** 84% of TSMs, 59.2% of technicians, and 90% of MMs reported waiting on parts multiple times a week or every day, suggesting parts delays are a regular constraint rather than an exception.
- **Perceptions of tool scarcity:** 30% of General Managers, 20% of Maintenance Managers, 28% of TSMs, and 39.3% of technicians reported that their site lacked necessary tools or equipment multiple times a week or every day, indicating that tool issues tend to be site specific, rather than as system-wide issue.

Comments[3]

- “Diagnostic computers for troubleshooting/maintenance are decrepit..no coordination of diagnostic software and computers to ensure the service center has all the latest diagnostic capabilities while also retaining the capabilities for all the Legacy outdated

[3] Comments have been paraphrased and sanitized to remove sensitive details while preserving the intent of respondent feedback.

systems. Techs often have to take it upon themselves to purchase personal diagnostic laptops and find proper programs, cables, and drivers, etc…"

- "The right tools for the job have been a struggle to understand what our service center will provide and what they won't…There has been a trend lately where technicians are now pooling together to buy tools to share in order to perform certain jobs."
- "The tool room's tracking process is a guessing game. The description of the tool mentioned in the manual aren't always what we call tools here at the service center…"
- "We have about half of the tools we need when we need them. Many times, they are in the other tool crib because we only have one of a tool that is needed in opposite ends of the building."
- "Frequent reorganization of the hangar taking up to an hour or longer, occurring at a rate of several times a week (sometimes several times per day)."
- "The internet is a tool we use daily. Our internet here is slow and it constantly slows us down from doing and completing our work."
- "...waiting on parts is simply part of the job."

Invoice Accuracy Themes

Theme 1: Role Design & Capacity

At several sites, CSM and invoicing responsibilities are layered on top of already full leadership and coordination roles, without enough capacity or clear role boundaries to support careful, consistent invoice review. Individuals "wear multiple hats" – managing projects start to finish, interacting with customers, overseeing teams, and handling billing. In some locations, TSMs also perform CSM or invoicing duties, further stretching bandwidth.

In this context, invoice accuracy competes with urgent operational work, and CSMs report relying on invoice specialists or peers to catch mistakes they do not have time to look for themselves. As a result, even when people care about doing accurate work, the way roles and workloads are designed makes thorough, systematic invoice review difficult to achieve in practice.

Evidence

Survey results

- Only 15% of CSMs agree that their team is adequately staffed and they have time to review invoices for accuracy
- Just 10% agree their workload is reasonable

Comments[4]

- "We are understaffed, and people work long hours to keep operations moving."
- "I rarely have time to review invoices thoroughly and often rely on others to catch mistakes."
- "Clear role boundaries are needed so CSMs aren't stretched too thin."
- "I manage projects from start to finish, interact with customers, oversee pilots, and handle billing—I wear multiple hats."

Cause Themes Impacting Both Problems

Theme 1: Process and SOP Standardization

Across service centers, core maintenance and invoicing activities are often carried out through locally defined processes rather than consistent organizational SOPs. On the operations side, processes and maintenance project management practices vary across sites, managers and schedulers rely on local tools, tribal knowledge, and personal judgment to plan, sequence, and manage work. Some teams have created their own spreadsheets or "best practice" documents, while others change processes when leadership changes. This variation means that basic elements such as job scoping, hour estimation, and shift handoffs differ from site to site, making cycle times and capacity planning heavily dependent on local workarounds and individual experience.

On the invoicing side, gaps exist. Standard processes for reviewing, coding, and completing invoices, are weak or absent, and documented guidance such as checklists or job aids are limited. CSMs report relying on peers, informal "cheat sheets," and the judgment of more experienced staff or the Invoice Review Team when they are unsure how to code work or verify coverage. Because each CSM develops their own way of working, key steps such as verifying coverage, checking data matches, and documenting decisions are applied inconsistently. This increases the likelihood that some invoices receive only a light review while others are checked more thoroughly, which contributes to avoidable errors, rework, and customer confusion.

Evidence

Survey results

- **Process clarity and consistency:** 52% of TSMs agreed their site's processes are standardized and consistently followed, but only 28% found those processes efficient. Among technicians, 52% agreed their site's processes are clearly defined and easy to

[4] Comments have been paraphrased and sanitized to remove sensitive details while preserving the intent of respondent feedback.

follow, while only 31% saw consistency in how jobs are completed across teams or shifts.

- **Invoice process guidance:** Only 30% of CSMs agreed that invoice systems and processes are standardized and easy to complete. Only 40% of CMs reported having access to a checklist for reviewing invoices, and just 25% had access to clear job aids to support accurate billing and data matches.

Comments[5] – Operations

- “Processes at my site are erratic and not consistent. It is getting better but with our management team, people seem to come and go and every time we get a new management team member we get a new process.”
- “One of the biggest challenges for scheduling is capacity… The scheduler and I have developed an internal spreadsheet for opportunities, this seems to ease scheduling requests and double booking.”
- “I selected mostly neutral or agree as LMNOP Aviation has a defined process in place for most technical issues and if there is not, we locally document a best practice to standardize our team with.”
- “Scheduling is primarily done by tribal knowledge. Our team gets the leads and TSMs in a room to estimate hours and due to their experience, they can pretty accurately estimate hours but there is limited context around their reasoning behind it so there can be worries of hoping it's correct sometimes.”

Comments[6] – Invoicing

- “There are no support processes to verify service coverage.”
- “The process is to use the internal teams to review.They try, but they lack detailed invoicing experience and can only offer suggestions when a definitive answer is needed.”
- “It [invoicing process] is kind of a standard but I think each CSM works on invoicing differently.”
- “[Invoicing review is] more of peer review in a sense where if you are not sure, we reach out to someone with more experience to confirm or reject prior to submitting.”

[5] Comments have been paraphrased and sanitized to remove sensitive details while preserving the intent of respondent feedback.
[6] Comments have been paraphrased and sanitized to remove sensitive details while preserving the intent of respondent feedback.

- “We can utilize the internal specialist, but you can only trust a few of them due to competency.”
- “...somebody will have a copy of, like, a cheat sheet. I'm saying, hey, can you send me that?”

Theme 2: Information and communication management

Across centers, managing work is challenging because critical information is fragmented across multiple systems and informal channels. Technical data needed to perform maintenance is scattered across portals with no simple, standardized access path. Communication between cross-functional teams and shifts relies heavily on email, texts, phone calls, Teams messages, and even sticky notes, with no designated system of record or standard workflow. As a result, staff spend time hunting for information, miss important updates (for example, new maintenance discrepancies or revised departure dates), and must piece together what was done on a job, undermining both productivity and invoice accuracy.

Similar fragmentation shows up in technical documentation. While manuals are generally viewed as accurate and available, in practice they are also scattered, requiring staff to search multiple sources. Communication and change updates follow the same pattern, with new procedures often introduced with minimal notice. Discrepancies between roles further highlight breakdowns across functions and frontline teams.

The absence of a consistent cross-functional communication workflow or system of record means staff can eventually find information, but only through time-consuming searches and invoice inaccuracies.

Evidence

Survey results

- **Communication processes:** 68% of TSMs and 50% of CSMs agreed that their site follows a standard communication process that keeps them informed.
- **Access to information and updates:** 70% of MMs agreed that technical documentation is accurate, up to date, and easily accessible. However, only 50% of MMs and 52% of TSMs agreed that updates and training on new or updated processes are timely, and just 36.5% of technicians agreed that process-related changes are clearly explained and communicated in a timely manner.
- **Data for invoicing:** Only 40% of CSMs agreed that they have access to reliable and accurate data from support teams necessary for accurate invoicing.

Comments[7] – Operations

- “LMNOP Aviation is the best at NOT communicating and launching new processes without any communication. When there is communication it is usually too late.”
- “Our online maintenance manual system changed overnight with no explanation and we had to figure out how to find them again.”
- “There is no workflow management program that is a single source to encapsulate the event. We rely on emails to communicate internally and externally with our customers, but that information is not memorialized in the event package or even available to the technicians working the event.”
- “We have information spread everywhere…Different OEMs store data in different places, and it can be very difficult for the team to find the information they need across multiple systems and websites…”
- “Communication at my site follows no set process at all… Handoffs seem to be at the discretion of who's writing it – sometimes it's printed on the classic handoff Sheet, sometimes it's typed up in a Word document, sometimes it's a sticky note that fell on the floor.”
- “We have terrible communication between shifts...”
- “Communication does not follow a standard process. When aircraft departure dates change by numerous days I am not informed and the scheduling system is rarely updated to add the new dates… When new maintenance discrepancies are added nobody marks them as AVIONICS so daily I have to check every work order to find out if an avionics Discrepancy was added.”

Comments [8]– Invoicing

- “We need to work on communication, especially with off shifts (customer comms) and handoff notes.”
- “It's [communication between CSMs/TSMs] mostly through phone calls and texts. There's emails also. It's kind of… it's just a combination of all three. There's also IMs.”

Theme 3: Digital Tools and System Design

Across service centers, the shift to “paper free” operations has increased reliance on digital tools and systems, but many of these tools are experienced as clunky, manual, and hard to

[7] Comments have been paraphrased and sanitized to remove sensitive details while preserving the intent of respondent feedback.
[8] Comments have been paraphrased and sanitized to remove sensitive details while preserving the intent of respondent feedback.

navigate. Technicians and CSMs describe excessive data entry, slow document load times, cumbersome workflows for tasks like clocking, parts removal, and task lookups. Validation to prevent errors is limited, and off-shift teams are particularly affected. When issues arise at night, staff often “either figure it out” or “it will have to wait for dayshift,” because technical or system support is unavailable. As a result, core systems that should streamline work instead add friction and delay, impacting both productivity and invoice accuracy.

Tool changes also tend to be rolled out with minimal communication or training, often “without much of an explanation or warning,” leaving employees to self-teach through trial-and-error, personal notes, and local workbooks. Training on technology is inconsistent across sites, and while job aids exist in some cases, many employees are unaware of them. This uneven support, combined with varying levels of technological proficiency, driving a heavy reliance on local workarounds (for example, alternative ways of entering parts when the “correct” process is broken). Configuration differences further amplify variation. At some sites, features like open/close lines and clocking are seen as “black boxes” that require manual policing to avoid time overruns, while other sites use the same tools to get clear, real-time visibility into hours by project and individual.

These design and configuration gaps are especially visible in tasks that depend on accurate, up-to-date system data.Verifying warranty or maintenance service coverage, for example, is often slow and cumbersome because systems do not reliably reflect current coverage, forcing staff to do extra legwork before billing. Non–user-friendly interfaces, manual processes, inconsistent training, and uneven configuration create avoidable delays, rework, and frustration. They also reduce trust in the systems themselves, making it harder for LMNOP to consistently meet productivity targets and maintain invoice accuracy.

Evidence

Survey results

- **Support for accurate billing:** Only 25% of CSMs agreed that invoicing systems and processes support accurate billing and data matches.
- **Timeliness of updates and training:** 50% of MMs, 52% of TSMs, and 35% of CSMs agreed that updates and training on new or updated processes/technologies are timely.

Comments[9] - Operations

- “The current Work Order system places a significant burden on technicians by requiring excessive data entry, particularly for parts, which detracts from their core responsibilities. It also lacks the flexibility for leads to organize projects effectively and does not facilitate clear communication between key stakeholders such as TSM, Avionics, Paint, Interior,

[9] Comments have been paraphrased and sanitized to remove sensitive details while preserving the intent of respondent feedback.

Quality, Parts, and Warranty. These limitations hinder both productivity and collaboration.”

- “...clocking and system is not user friendly. The system requires too many manual functions and coding to the point where mistakes are often made. The only reason we feel comfortable with the system now, is because we have been working it for years and have made so many errors and have learned from our mistakes.”
- “...our maintenance team is extremely frustrated with the work order platform. Removing paper processes has been the latest hurdle, creating significant delays just to locate a removal. We’ve already started using workarounds to enter parts because the correct process is broken.”
- “The recent transition to the online manual system has notably increased document load times, doubling the time it takes to access necessary information.”
- “...we've recently gone paperless. And our software program… we're constantly going through changes with that. And it results in a lot of interruptions.”
- “System is tedious to work within…Since shifting reporting metrics to Power BI, the data I'm looking for is harder to find. I tend to look at system generated reports rather than the BI's.”
- “Parts often has no stock of medium to high value parts and gives no notification that a part must be allocated, often delaying the order process several days because there is no notification the part went into limbo.”
- “The teams that make all of the IT/Training changes roll out updates without much of an explanation or warning.”
- “Was there any sort of formalized training on the technology or software or processes? No. I make notes in it on how to do certain things, but there are a few, workbooks, that are pretty clear on step-by-step on how to do stuff.”
- “Part of the new program, is my… issue is the open and close line. They are hard to regulate. Who gets on that Discrepancy? Who puts time on it? And it usually puts us over on our time.”
- “Our system also tells us hours, a live update of... how many hours are on a project, it tells us who built it and how many hours each individual built it…”

Comments [10]- Invoicing

[10] Comments have been paraphrased and sanitized to remove sensitive details while preserving the intent of respondent feedback.

- “Less complex billing system, one that is intuitive and not a labyrinth of options and avenues that I will never use in my role” (in response to what would help them do their job more efficiently).
- “Some items do not code automatically. Also, there are time rates missing on some models. The Power BI is not easy to search and find what you need”
- “Quotes for Mods sometimes break and they must be manually matched to the proposal. I've invoiced a Revision 32 times in an attempt to match the proposal.”
- “I think the system we use works as intended, but it is not straight forward and I cannot trust technical codes or Coding of the Discrepancy at times.”

Theme 4: Training, onboarding, and capability systems for key roles

Across the network, there is no consistent, organization-wide approach to building and maintaining capability for technicians or CSMs. New and less-experienced technicians usually ramp up through informal, on-the-job learning, shadowing, and “tribal knowledge,” with limited structured cross-training or clear expectations of each grade level's skill sets. Technician grades are based primarily on tenure and certifications rather than defined skill sets, making it difficult for TSMs to plan work and assign tasks confidently and leaves technicians frustrated about how to progress.

Productivity is increasingly constrained by the loss of experienced technicians and greater reliance on newer staff in an industry already facing shortages of qualified mechanics. Scheduling norms place many of the least experienced technicians on weekend or night shifts while senior staff work more desirable day shifts, weakening real-time coaching and creating gaps in knowledge transfer. Crews are not consistently cross-trained across aircraft types, manuals vary in quality and usability, and newer technicians often depend heavily on the informal guidance of older peers. While some centers promote mentoring and pairing new hires with experienced mechanics, progression depends heavily on individual leads and local site culture. Combined with external labor pressures, these factors (tenure-based grading, uneven cross-training, weak knowledge-transfer systems, and off-shift staffing patterns) slow ramp-up, reducing confidence among newer staff, reducing productivity.

On the invoicing side, the pattern is similar but more pronounced. There is no standardized onboarding or training curriculum for CSMs or invoice-related roles, and many CSMs are hired or promoted primarily for interpersonal and coordination skills rather than aviation expertise. Most describe “learning as you go” through trial-and-error, ad hoc explanations from peers, and scattered references. Key tasks, such as interpreting technical reason codes, understanding coverage, and translating complex maintenance work into accurate invoices, lack clear job aids and structured practice, leading to coding errors, rework, and customer perceptions of CSMs as “financial people” rather than trusted OEM advisors. When CSMs are unsure how to code work or understand the work performed, questions are pushed back onto TSMs, adding to their workload and creating additional handoffs. Gaps in training, onboarding, competency definition, and knowledge-transfer systems for both technicians and CSMs create wide variation in capability and negatively impact productivity and invoice accuracy.

Evidence

Survey results

- **Technician availability:** 60% of GMs, 10% of MMs, and 52% of TSMs indicated that insufficient technician availability occurs at their site multiple times a week or every day, highlighting perceived gaps in experienced capacity.
- **Adequate training – leadership vs frontline views:** 80% of GMs and 70% of MMs indicated that employees receive the training they need to perform their roles effectively. Among frontline roles, 65.1% of technicians agreed they had received the training needed for their role, while only 30% of CSMs agreed they had received the training needed, even though 60% felt trained to identify and correct invoice errors. 54.4% of technicians agreed there is ongoing training available to accommodate changes in systems and processes.
- **Performance support and job aids:** 87.7% of technicians reported knowing where to go for help or guidance when needed, but only 25% of CSMs indicated they have access to clear job aids to support accurate billing and data matching.
- **Skills development:** 55.1% of technicians and 55% of CSMs agreed there are opportunities to develop their skills and advance.

Comments[11] – Operations

- "Finding good, experienced technicians because that is what we are lacking the most."
- "Our team is relatively new to the industry, which makes it challenging for them to anticipate and proactively address potential issues. To support their development, we rely on the tools provided by the company and encourage collaboration across the team."
- "When looking for information I'm often told to 'just read the manual'. I do as I'm told but sometimes I find out that there is some tribal knowledge for example, there's a tool or adapter that needs to be used that the manual doesn't call out for. My experience so far has been do the job incorrectly as in break something or take too long. That's when time is given to technicians to properly train them."
- "Initial training was almost non-existent. I have had to learn what my job was over time. After a couple years I am still finding out areas of my job that I should be taking care of."

[11] Comments have been paraphrased and sanitized to remove sensitive details while preserving the intent of respondent feedback.

- “We normally pair them up with a more experienced technician… Each person kind of progresses a little bit different, at a different pace.”
- “Training needs to be standardized for all grades by grade so we know what to assign and be able to hold the technicians accountable.”
- “The current scheduling system does not factor into consideration skillset only tech capacity which creates additional challenges when scheduling.”

Interview comments[12] – Invoicing

- “It would be nice to have some detailed invoice training for new CSMs.”
- “Knowledge of the invoicing process has been learned from hands on experience.”
- “…everything I have learned for my roles has been on the job learning from another CSM.”
- “I did not receive formal training. CSMs were too busy to be able to just sit down and explain processes. The only way for me to gain experience was to just go ahead and start owning projects (trial by fire).”
- “Is there a standardized, like, process to get the CSMs who don't have a TSM background [technical] information on the process... Yeah, not necessarily, yeah, you just kind of learn as you go on.”
- “My team sometimes misses minor details that become pain points with customers. I see it primarily in my CSMs as I've gotten customer feedback that they feel like their CSMs are just financial people, and wished they were advisors..”

Theme 5: Performance management and culture

Performance management and accountability vary widely across the network,with frontline experience depending heavily on the site and immediate leadership. Leaders generally report that expectations are clear and that high performance is modeled and reinforced, but technicians and CSMs describe a more mixed reality: expectations can feel implicit or shifting, constructive feedback is inconsistent or mostly negative, and underperformance is not always addressed while strong performance is treated as “just doing your job.” Many technicians also report getting little guidance on how to progress to higher grades or what “good” looks like beyond hitting basic metrics.

Pay and progression structures compound this issue. In some cases, supervisors who moved up from lead technician roles now earn less than they did previously, or less than the

[12] Comments have been paraphrased and sanitized to remove sensitive details while preserving the intent of respondent feedback.

technicians they manage. This pay compression undermines the message that leadership and performance management are valued, and it discourages high-performing technicians from moving into TSM, CSM, or supervisor roles. Together, inconsistent feedback and accountability practices, tolerance of underperformance, and weak incentives to step into leadership roles limit the organization's ability to build and sustain a performance-focused culture that will improve invoice accuracy and productivity.

Finally, employees at multiple levels describe a strong emphasis on hitting metrics without equivalent support for coaching or problem-solving. Productivity and invoice accuracy numbers are tracked, but feedback often focuses on whether targets were met rather than on understanding root causes or reinforcing effective behaviors. This emphasis on output over outcomes makes it harder for employees to learn from issues, feel recognized for quality work, and see how their efforts contribute to long-term performance.

Evidence

Survey results

- **Modeling and accountability:** 80% of GMs and 90% of MMs indicated that high performance and accountability are modeled and reinforced by leadership at their site. This drops to 64% for TSMs, 60% for CSMs, and 48.5% for technicians. 60% of CSMs indicated that when invoicing errors occur, their leadership responds fairly and with appropriate accountability.
- **Feedback and culture:** 60% of GMs and 90% of MMs indicated that constructive feedback and recognition are part of their site's culture. 68.3% of technicians reported feeling comfortable sharing ideas or concerns with their supervisors.
- **Clear expectations:** 90% of GMs and 100% of MMs indicated that they have a clear and shared understanding of job expectations for their teams. 92% of TSMs reported a clear understanding of what defines a "high quality job," and 90% of CSMs indicated that expectations for invoice accuracy are clearly defined and communicated. 69.9% of technicians agreed that their supervisor communicates performance expectations clearly.
- **Receiving feedback:** 75% of CSMs indicated that feedback on invoice accuracy from their leadership is helpful and timely. In contrast, 56.3% of technicians agreed that they receive feedback that helps them improve their performance, and only 45.2% indicated that they receive consistent feedback about their ability to achieve productivity goals.

Comments[13] - Operations

- "They are busy. we are busy. Not a lot of time to talk about performance."

[13] Comments have been paraphrased and sanitized to remove sensitive details while preserving the intent of respondent feedback.

- “A lot of the times I feel like I'm supposed to read minds or just figure out what needs to be done.”
- “I have asked for feedback and the answer has always been ‘Nothing I can think of,’ or the more usual ‘I don't know.’”
- “My supervisor typically communicates with negative feedback, rarely anything positive or constructive.”
- “Recognition for strong performance often translates into additional workload rather than meaningful acknowledgment or reward. Performance metrics and actual results appear to have limited influence on feedback, which tends to be negative regardless of outcomes. When the company performs well, the perception is that we could have done more; when performance declines, the responsibility falls squarely on our performance.”
- “Supervisors and leads sometimes get upset with mechanics but never provide any helpful feedback…Also, the clocking expectations are always changing… Management comes once a year and says to clock truthfully even if there's no work, but then the supervisor will always get mad at you saying you are clocking to DCI too much…”
- “I never receive feedback from my supervisors that is constructive, instead I feel that all feedback I get is limited to mainly criticism and produces an environment where I feel my ability to succeed is severely limited, a feeling that was amplified when I asked my supervisor how I can move up to a Grade A and his response was ‘Uh, I don't know.’”
- “My coworkers underperform without any consequence. It's disheartening to work hard when there are those beside you that don't seem to care about making an effort.”
- “I do feel as though my hard work is just viewed as ‘doing my job’. When someone is slacking however, they get the same treatment as someone that is going above and beyond.”
- “...some supervisors who previously served as lead technicians now earn way less than their former role. This disparity discourages career progression and contradicts the intent of promoting leadership opportunities.”

Comments[14] - Invoicing

- “Our new GM does a phenomenal job holding people accountable and setting expectations, however, our TSM level needs some improvement on this.”
- “Equal distribution of work and holding each CSM accountable for their work is imperative.”

[14] Comments have been paraphrased and sanitized to remove sensitive details while preserving the intent of respondent feedback.

- "Leadership is quick to point out if an error is made. I don't think they do a very good job of holding people accountable or with recognition of a job well done."

Intervention Selection/Feasibility

Selecting interventions to address performance gaps requires a structured approach that balances impact with practicality. In this phase, Team Aviation applied strategic analysis tools to ensure recommendations were both evidence-based and aligned with LMNOP Aviation's organizational goals. We began by synthesizing insights from the diagnostic phase into key themes, then used SWOT analysis to evaluate internal strengths and weaknesses alongside external opportunities and threats. This helped identify leverage points and constraints shaping LMNOP's ability to improve productivity and invoice accuracy. To further prioritize options, we later applied Multi-Criteria Analysis (MCA), assessing interventions against factors such as feasibility, cost, time to implement, and work stoppage. Together, these methods provided a rigorous foundation for selecting interventions most likely to deliver sustainable improvements.

SWOT Analysis

SWOT analysis is a strategic planning tool that evaluates four key dimensions of an organization or situation to inform decision making (Stefaniak, 2020). These four dimensions are:

- Strengths: the Internal positive attributes and capabilities that give the organization an advantage
- Weaknesses: the internal limitations or gaps that hinder performance
- Opportunities: the external or internal factors that the organization can leverage for improvement
- Threats: the external or internal factors that could negatively impact the organization.

Performing a SWOT analysis helped Team Aviation systematically evaluate LMNOP's current position and identify the root causes with the greatest impact. By understanding existing strengths to build on, weaknesses to address, opportunities to pursue, and threats to mitigate, we were able to focus on the causes most realistic and strategically aligned with LMNOP Aviation's organizational goals.The themes synthesize the causes of performance problems impacting both productivity and invoice accuracy that surfaced across data collection methods capturing what is happening now, why it matters, and where the leverage is. Details supporting SWOT analyses for each theme are provided in Appendix B. Together, these themes describe a system in which local processes, fragmented information, clunky tools, informal capability systems, and uneven performance management interact, driving the observed gaps in productivity and invoice accuracy.

Table 5

SWOT Cross-Theme Summary – Shared Causes Impacting Productivity and Invoice Accuracy

Theme	Core Observation	Key Strengths to Leverage	Key Weaknesses/Risks	Levers
Process and SOP Standardization	Inconsistent processes are preventing the network from fully converting paid labor into reliable, billable output and accurate invoices, despite having strong local examples of what “good” looks like.	Some sites have well-developed local practices and informal SOPs; experienced TSMs and CSMs know how to “make it work” and are already using spreadsheets, peer review, and local best practices.	Processes are person- and site-dependent; maintenance planning, handoffs, and invoice review lack consistent steps, checklists, and quality controls, making performance fragile and hard to compare across sites.	Design and roll out network-wide SOPs and checklists for maintenance project management and invoice review; use high-performing sites as prototypes; add light governance so standards are maintained and improved.
Information and Communication Management	Technical and operational information is fragmented across multiple systems and informal channels, with no single source of truth or standard communication	The information generally exists somewhere (systems, manuals, email, Teams), and teams have already created workarounds (e.g., checking every work order, local handoff practices), showing awareness and effort.	Manuals are spread across multiple sites and portals. No designated system of record for an event; cross-shift and cross-functional updates are inconsistent; changes to maintenance discrepancies, schedules, and programs are easily	Define a single system of record for each event and a standard communication/update workflow; create simple handoff templates; use automation (alerts/flags) and targeted training on “where to put what.” Create a centralized knowledge hub or portal where all

	workflow; staff spend time hunting, re-confirming, and sometimes miss updates.		missed, which slows work and drives invoice errors.	technical documentation is accessible.
Digital Tools and System Design	“Paper free” tools for maintenance and invoicing are often clunky, manual, and weakly validated, slowing users down and allowing incomplete or incorrect entries to pass through to invoices.	Core platforms are already in place, and staff have shown resilience and creativity in learning and using them despite friction; digital records can support better traceability once workflows are improved.	Poor usability (multiple screens, confusing fields), heavy manual entry, weak validation, and poorly communicated updates create friction, errors, and inconsistent use across sites.	Reconfigure tools around real workflows (UX review), strengthen validation and business rules, embed standard process steps in the systems, and improve change management and support for updates.
Training, Onboarding, and Capability Systems for Key Roles	Capability for both technicians and CSMs is built largely through informal, locally-driven learning rather than consistent, organization-wide training and competency systems. This creates wide	Strong informal support networks and mentoring at some sites; many techs feel they know where to go for help; OEM content and existing training resources can be built on; some CSMs bring strong customer/coordination skills.	No standardized curricula or grade-based competency expectations; off-shift staffing puts many newer techs without strong senior support; CSMs report “learning as you go” for complex coding and coverage decisions, driving errors and extra handoffs.	Define competency models and grade expectations; build structured onboarding and training for techs and CSMs; implement skills matrices; formalize mentoring

	variation in skills across sites and shifts, slows ramp-up, and increases reliance on a shrinking pool of experienced staff to keep maintenance moving and invoices accurate.			
Performance Management and Culture	Performance expectations and metrics are clear on paper, but day-to-day feedback, recognition, and accountability are uneven across sites; strong focus on hitting numbers often outweighs learning, problem-solving, and fair recognition.	Some sites already use performance management systems and localized goals well; many leaders value accountability; technicians often feel comfortable raising concerns; CSM invoice feedback is helpful in some locations.	Perception gap between leaders and frontline staff; feedback is often infrequent or mainly negative; underperformance is tolerated while strong performance feels unrecognized; pay compression makes TSM/supervisor roles less attractive for high-performing techs.	Define a simple performance-management standard for service centers; invest in frontline leader coaching skills; realign TSM/supervisor pay and career paths; rebalance metric reviews toward metrics + causes + behaviors; strengthen recognition practices and spread good site-level examples.

Possible Interventions

To address the performance gaps identified in the analysis phase, Team Aviation developed a set of target interventions grounded in the themes and leverage points that surfaced though our SWOT analysis. Each intervention is designed to mitigate one or more root causes impacting productivity and invoice accuracy.

Our selection process integrated multiple data sources to ensure recommendations were evidence-based and strategically aligned:

1. Quantitative Survey Insights: Structured responses were analyzed and categorized using the BEM and Nine-Variable framework to identify performance gaps and capability needs. Categories were prioritized based on the lowest average Likert scores, while also considering question density and respondent coverage to ensure reliability. For example, a category with only one question asked exclusively to CSMs was not elevated as a priority, since limited data made its average less representative.

   In Figure 6, we demonstrated our Likert scale results. They are visually highlighted in red, yellow, and green, indicating low, medium, and high scores respectively. This color-coding, paired with question count, guided our focus toward categories most critical for improvement. Using this approach, we identified priority areas within the BEM and Nine-Variable frameworks. These low-performing categories are outlined in green in Figure 6.

Figure 6

Survey Results Overview by Likert Avg Rating and Question Count

Chevalier's BEM

Environmental Factors	Information	***Avg Likert Score: 3.8 Question Count: 40***
	Resources	***Avg Likert Score: 2.9 Question Count: 24***
	Incentives	***Avg Likert Score: 3.2 Question Count: 11***
Individual Factors	Knowledge/Skills	***Avg Likert Score: 3.4 Question Count: 15***
	Capacity	***Avg Likert Score: 2.4 Question Count: 1***
	Motives	***Avg Likert Score: 3.9 Question Count: 20***

Rummler and Brache's 9 Boxes

Top Row: Organizational Level	Organization Goals	***Avg Likert Score: 4.9 Question Count: 3***
	Organization Design	***Avg Likert Score: 4.2 Question Count: 2***
	Organization Manag	***Avg Likert Score: 4.6 Question Count: 1***
Individual Factors	Process Goals	***Avg Likert Score: 3.5 Question Count: 15***
	Process Design	***Avg Likert Score: 3.4 Question Count: 8***
	Process Manageme	***Avg Likert Score: 3.3 Question Count: 25***

2. Qualitative Survey Insights: Open-ended survey comments and interview narratives provided context behind the numbers from the survey ratings, revealing systemic issues and reinforcing key themes from the SWOT analysis.

By combining these quantitative and qualitative inputs, we identified priority areas for intervention. To ensure alignment and effectiveness, we applied Hale's Intervention Family Framework Table 6, which classifies organizational interventions by purpose and style. This framework supported evidence-based selection of solutions tailored to LMNOP's organizational needs and readiness for change.

Table 6
Full List of Interventions Considered

BEM/Nine-Variable Model Category	Hale's Intervention Family	Intervention
Process Design	*Standardize*	**Standardize processes and operating procedures** Capture best practices from top-performing centers and formalize them into standardized workflows for planning, invoicing, communication, and tool management.
Process Management	*Measure*	**Centralized KPI dashboard** Develop a single dashboard that displays key metrics: productivity, invoicing accuracy, and financial performance, for real-time visibility and decision-making.
	Standardize	**Standardized escalation protocols** Create a formal plan for escalating issues during maintenance events to ensure problems are addressed early, preventing delays and missed deadlines.
Knowledge	*Develop*	**Role-specific onboarding paths** Build standardized onboarding and training curricula for key roles, including structured practice with real scenarios and system use.
	Develop	**Formal Mentorship Program** Formalize and scale mentoring practices to support knowledge transfer through structured pairing and peer learning.
	Develop	**Bridge shifts** Schedule 2-3 hours of overlap between shifts.
	Reframe/Redefine	**Rotating KSA Champions** Develop champion roles at each site to support peers, surface usability or process issues, and share best practices across sites.
	Develop	**Simulation based training**

		Create technical training using AR or VR to help technicians build maintenance skills without waiting for specific aircraft availability.
	Develop	**Peer learning shadowing** Offer role shadowing opportunities to allow employees to gain experience in other functions.
	Develop	**Frontline leadership development** Invest in leadership development for TSMs and leads, focusing on practical skills for coaching, goal setting, and performance management.
	Develop	**Targeted training** Offer focused training on optimized workflows and digital tools, including proper use of the system of record and key application features.
Information	*Define*	**Competency Model** Define competency models and skill expectations for key roles to guide work allocation, feedback, development, and advancement.
	Document	**Job Aids** Create job aids and document SOPs for critical tasks, including invoicing, digital tools, maintenance procedures, and communication workflows.
	Inform	**Regular one-on-one meetings** Establish a regular cadence for one-on-one meetings (weekly or monthly depending on role). Topics include feedback, development, performance.
	Define	Skills matrices Implement skills matrices at site and network level to give TSMs clear visibility into who can do what and to plan cross-training deliberately.

	Inform	Mobile app with live work updates Mobile app work dashboard for TSMs and Leads to keep track of who is working what live
	Design/Document	Change management for system updates Improve (or implement) change management for system updates (advance notice, quick-reference guides)
Incentives	*Reward*	Strengthen recognition programs for excellence Enhance recognition practices to reward quality work, effective coaching, and thorough invoice review, while promoting awareness of LMNOP's program and sharing best practices across sites.
	Reward	**Leadership compensation review** Review and adjust pay bands and career paths so moving into leadership is financially and professionally attractive for strong technicians
Resources	*Redesign*	**AI powered invoicing** Create or buy an AI agent or program to handle the bulk of creating and completing invoices.
	Standardize	**Create and embed handoff templates** Create simple, consistent shift-handoff templates and embed them in existing tools
	Standardize	**Single system of record** Define and implement a single system of record for each maintenance event (even if underlying systems remain separate), with clear rules about what must be captured there.
	Document/Redesign	**Centralized information hub (RAG LLM or Knowledge Hub)** Create a central information hub for manuals, job aids, sharepoints, and other necessary documentation. Create a RAG LLM to make locating information fast and easy.

	Measure/Redesign	**Conduct a system and programs assessment** Evaluate maintenance and invoicing systems to simplify workflows, automate alerts, and strengthen validation rules for greater efficiency and accuracy.
	Redesign	**Update outdated tools** Review and correct tool issues at affected locations.
	Develop	**Wifi connectivity** Explore options for upgrading wifi at affected locations.
	Organize	**Parts racks** Provide additional part racks to prevent misplaced components
	Organize	**Sheds** Add tool storage sheds to improve organization and accessibility
	Redesign	**Facility capacity and size** Assess hangar layouts and capacity at all locations, and explore options for reconfiguration, expansion, or relocation to eliminate bottlenecks and outdoor work.

Multicriteria Analysis

After identifying a comprehensive set of interventions through Hale's Intervention Family Framework and tying potential interventions back to BEM and Nine-Variable Model, Team Aviation needed a systematic way to determine which options would deliver the greatest impact with the least disruption. To move from possibilities to priorities, we applied a Multi-Criteria Analysis (MCA). This approach allowed us to evaluate interventions against LMNOP Aviation's strategic goals and operational constraints, ensuring that recommendations were both feasible and high-value. The following section outlines the MCA process, criteria used, and how these informed our final selection of interventions.

Team Aviation employed the multicriteria analysis to systematically evaluate and prioritize interventions that would most effectively address identified root causes. Multicriteria is a decision-making tool that compares multiple options by assigning weighted numeric values to each based on defined criteria (Watkins et al., 2011). This structured approach moves beyond identifying problems to recommending evidence based solutions with the greatest organizational impact.

The team collaborated with LMNOP Aviation's general managers to define decision criteria aligned with strategic goals and operational realities Table 7.

Table 7
MCA Criteria and Guiding Questions

Criterion	Scale	Guiding Question
Cost	5 = Low cost, 1 = High cost	How expensive would it be to develop and implement the intervention?
Work Stoppage	5 = No disruption, 1 = Complete stoppage	Would implementing the intervention interrupt normal operations?
Speed of Implementation & Results	5 = Noticeable results within months, 1 = Minimal impact	How quickly can the intervention be implemented and deliver measurable outcomes?
Sustainability	5 = Highly sustainable, 1 = Difficult to maintain	Is the intervention easy to maintain over time?
Impact to Performance Metrics	5 = Strong improvement in productivity and invoice accuracy, 1 = No impact	Does this intervention improve invoice accuracy, productivity, or both?
Number of Locations Affected	5 = Network-wide, 1 = Limited scope	Does this intervention have a large impact across the network?

Each potential intervention was rated on these criteria, weighted by importance, and scored to produce a quantitative comparison. This process ensured prioritization was data-driven, transparent, and aligned with LMNOP Aviation's operational priorities, Table 8.

We've summarized the Multi-Criteria analysis below. The full analysis is in Appendix B.

Table 8
Summary of Results Multi-Criteria Analysis

Commented [1]: Come back to this

Intervention	**Cause Theme**	**Final Rating**

Process Design		
Standardize Operating Procedures	Process and SOP Standardization	4.25
Process Management		
Create a centralized KPI dashboard	Digital Tools and System Design	3.35
Develop standardized escalation protocols	Process and SOP Standardization	3.7
Knowledge and Training		
Role Specific Onboarding Paths	Training, onboarding, and capability systems for key roles	4.3
Formal Mentorship Program	Training, onboarding, and capability systems for key roles	4.15
Bridge shifts	Training, onboarding, and capability systems for key roles	3.4
Rotating KSA Champions	Training, onboarding, and capability systems for key roles	4.55
Simulation based training	Training, onboarding, and capability systems for key roles	2.85
Peer learning shadowing	Training, onboarding, and capability systems for key roles	4.2
Frontline leadership development	Performance management and culture	4
Targeted training	Process and SOP Standardization	4.3

Information		
Competency Model	Process and SOP Standardization	4.1
Job Aids	Process and SOP Standardization	4.5
Regular one-on-one meetings	Performance management and culture	4.1
Skills matrices	Process and SOP Standardization	4.65
Mobile app with live work updates	Information and communication management	3.2
Change management for system updates	Digital Tools and System Design	4.5
Motives		
Career pathways leadership development program	Performance management and culture	4.65
Incentives		
Strengthen Recognition programs	Performance management and culture	4.55
Leadership compensation review	Performance management and culture	4.35
Resources		
AI powered invoicing	Role Design & Capacity	3.2
Centralized information hub (RAG LLM or Knowledge Hub)	Information and communication management	4.25
Create and embed handoff templates	Information and communication management	4.66

Conduct system and programs assessment	Digital Tools and System Design	4.05
Single system of record	Digital Tools and System Design	4.67
Update outdated tools	Physical environment & equipment	3.5
Wifi connectivity	Physical environment & equipment	3.75
Parts racks	Physical environment & equipment	3.7
Sheds	Physical environment & equipment	3.7
facility capacity and size	Physical environment & equipment	3.5

Table 9 lists interventions that were not included in Team Aviation's recommendations. These interventions were items either suggested by leadership and employees, or part of the potential interventions evaluated by the multi-criteria analysis. The table includes the potential interventions and the reason it was not selected.

Table 9
Rejected Interventions

Intervention	**Reason for Rejecting**
Project Management Training	Does not address systemic issues of uneven processes across sites. Can not train on processes that are not standardized.
New Computer Software Purchases	Replacement of the system would incur a high cost and create an extensive implementation timeline without guarantee of improvement.
Staffing Increases	SOPs should be established and improvements noted before large-scale staffing shifts.

Return to Paper Processes	Creates delays, risks data loss, and does not allow for real time updates. It also could increase invoice error rates and lead to documentation complications.
Technical Client Manager Role	Would create inconsistency in client communication and additional cost for service centers to create this role. Other recommended interventions may alleviate the need for this role.
Building additional/larger hangars	Potentially valuable, but site-specific and not enough information to justify inclusion.
Create a centralized KPI dashboard	Although it's quick, low cost, and relatively simple to implement, it's not likely to have a high impact on productivity metrics. Other recommended interventions, such as regular 1-1s, will be more impactful.
Bridge shifts	While it's quick, low-cost, and easy to implement, it's unlikely to meaningfully improve productivity metrics. Other interventions, like regular 1:1s, are expected to have greater impact.
Simulation-based training	This option would enable more controlled skill development, but it's not financially feasible right now. The required equipment and the cost of hiring vendors or internal AR/VR development expertise are prohibitive.
Mobile App with Live Work Updates	Potential implementation cost and redundancy with other application tracking tools make the investment less desired.
AI Invoicing Agent	Potential implementation cost and potential error rates make this option high risk. However, as agents improve it may be worth reevaluating in the future.

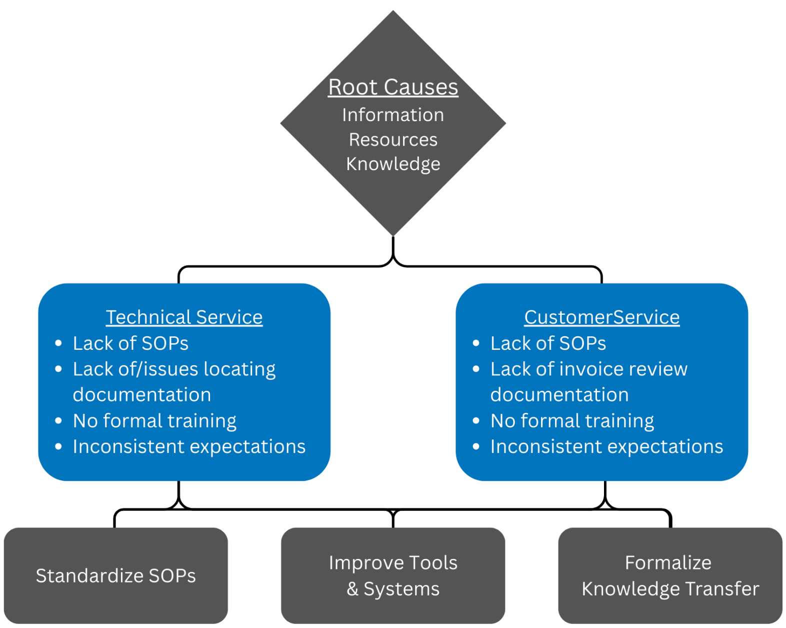
Intervention Recommendations

Team Aviation identified several interventions that address the systemic root causes of productivity and invoice accuracy underperformance across LMNOP Aviation's domestic service center network, shown in Figure 7. Our analysis indicates performance problems primarily stem from organization-wide factors rather than individual performer issues.

The recommended interventions focus on information, resources, and knowledge, which correspond to three of the lowest-scoring categories in our survey analysis. By targeting these areas, the interventions directly support the BEM framework and are designed to drive measurable, sustainable results.

Figure 7

Shared root causes and primary recommendations for Productivity and Invoice Accuracy



Because the proposed interventions address root causes impacting both productivity and invoice accuracy, our recommendations are organized into three tiers:

- Primary Recommendations – Network-wide interventions targeting key factors that influence productivity and invoice accuracy across all locations.
- Second-Level Recommendations – Network-wide interventions that support overall performance but are more complex to implement, may yield lower ROI, or require greater resources than first-level priorities.
- Third-Level Recommendations – Site-specific infrastructure improvements that enhance productivity, capacity, and turnaround time at affected locations.

Primary Recommendations

Based on triangulated data, root cause analysis, and established selection criteria, three primary recommendations were developed to address systemic factors impacting productivity and invoice accuracy. These recommendations were prioritized for their potential to deliver the greatest benefit across the network relative to cost and implementation complexity and focusing on our identified frameworks Chevalier's BEM and Rummler and Brache's Nine-Variable Model.

- Standardize Processes (SOPs) - ***Information/Knowledge/Process Design***
- Improve Tools and Systems - ***Information/Resources/Process Management***
- Formalize Knowledge Transfer - ***Information/Motivation/Resources***

Standardize Processes (SOPs)

For the first recommendation, it is proposed that LMNOP Aviation utilize existing working groups or designated process leaders to identify and document best practices from exemplary sites. These standards should be established across the following areas:

- Project planning, scheduling, and management
- Invoice review and quality control
- Work allocation and assignments
- Capability model for technician levels and job descriptions
- Project communication (who records what, where, and when) for maintenance discrepancies, schedule changes, service coverage decisions, and handoffs between shifts.
- Tool maintenance and tracking
- Create job aids and document SOPs for the following tasks:
    - Invoicing process (fault-code guides, service coverage decision trees, invoice review checklists, etc).
    - Digital tools
    - Maintenance processes and procedures
    - Communication workflow/single system of record
- Create and embed handoff templates
- Create a skills matrix based on the competency model to help with capacity planning

By formalizing and documenting these practices and developing work standards, LMNOP Aviation can promote consistency across operations, enhance efficiency, and ensure organizational knowledge is effectively captured and applied throughout the network.

Improve Tools and Systems

For the second recommendation, it is proposed that LMNOP Aviation prioritize improvements to its tools and systems through a series of targeted redesigns. These efforts are intended to streamline processes, enhance reliability, and ensure that critical information is consistently accessible across the organization. The following actions are advised:

- Centralize and normalize documentation use (Knowledge Hub, KMS, RAG LLM)
- Define and implement a single system of record for each maintenance event (even if underlying systems remain separate), with clear rules about what must be captured there
- Conduct assessment of system and other tools to understand where they can be improved
- Improve system performance and reliability
- Improve (or implement) change management for system updates (advance notice, quick-reference guides)

By implementing these measures, LMNOP Aviation can strengthen its technological infrastructure, reduce inefficiencies, and create a more resilient framework to support future growth and innovation.

Formalize Knowledge Transfer

For the final primary recommendation, it is proposed that LMNOP Aviation implement initiatives designed to enable experienced technicians to foster and develop newer staff. Establishing this pathway will help preserve critical tribal knowledge that often departs with retirement, while also accelerating the learning curve and capability development of new hires. The following actions are advised:

- Formalize and scale mentoring/knowledge-transfer practices (e.g., structured pairing, off-shift coverage by experienced techs, communities of practice).
- Build standardized onboarding and training curricula for key roles (techs and CSMs), including structured practice on real scenarios and use of systems
- Certification pathway tied to grade levels based on job duties

By institutionalizing these practices, LMNOP Aviation can ensure that organizational expertise is retained, professional growth is supported, and workforce readiness is strengthened across all sites.

Second Level Recommendations

Second level recommendations represent initiatives that still address widespread challenges across multiple facilities but are more complex to implement, might have less ROI, or require greater resources than first-level priorities. These recommendations are designed to strengthen organizational capability and long-term resilience rather than deliver immediate operational fixes. While this might all be true these recommendations impact would still be significant to the organization based on our findings. As with the primary recommendations these focus on three areas in Chevalier's BEM model and Rummler and Brache's Nine-Variable Model.

- Strengthen Performance Management - ***Motives/Knowledge/Information***
- Develop Career Pathways - ***Motives/Incentives/Resources***
- Enable Peer Learning - ***Knowledges/Motives/Process Management***

By implementing these second level recommendations, LMNOP Aviation can expect meaningful improvements in workforce stability, leadership and employee accountability and development, and knowledge retention.

Strengthen Performance Management

For this second level recommendation, it is recommended that LMNOP Aviation strengthen performance management. While this initiative may require more resources than the first level priorities, its impact on accountability and workforce engagement will be significant, The following actions are advised to improve performance management and better service the workforce of LMNOP Aviation:

- Train managers on delivering effective, consistent feedback
- Implement regular feedback patterns, such as weekly or monthly one-on-one meetings
- Strengthen recognition programs to reward excellence and reinforce desired behaviors. Promote the LMNOP recognition program.

By institutionalizing these practices, LMNOP Aviation can create clearer expectations, accelerate performance improvement, and foster a culture of accountability. Although this recommendation is more complex to implement, it aligns with Chevalier's BEM and Rummler and Brache's Nine-Variable Model by addressing gaps in motives, knowledge, and information, but also in organizational processes. The feasibility of this intervention is high because of the minimal cost and time constraints, but also because the long-term benefits will include stronger leadership capability and sustainability.

Develop Career Pathways

This second level recommendation is to implement career pathways These pathways will give individuals a route to leadership and growth. The goal of these pathways is to strengthen leadership capability, accountability, and reduce turnover. While this initiative may require more resources and planning than the first level priorities, its long-term impact on retention and organizational resilience will be significant. WIth these career pathways there needs to be incentives and an adjustment in the mindset of professional growth. For this suggested intervention the following actions would be recommended:

- Adjust leadership compensation and review cost of living differences across locations to ensure equity and make leadership roles more attractive
- Create leadership development program for aspiring managers to build a stronger pipeline of qualified candidates
- Establish clear competency requirements for advancement so employees understand expectations and growth opportunities

The expected impact includes reduced employee turnover and increased motivation for employees to join and remain in leadership positions. The feasibility of this implementation varies based on how it is approached but is generally rated as medium to high depending on the organizational departments and processes involved. By implementing this solution LMNOP Aviation can create a culture of growth and opportunity, ensuring long-term stability and strong leadership.

Enable Peer Learning

For the final second level recommendation we advise that LMNOP Aviation enables peer learning to foster collaboration, share best practices and strengthen organizational knowledge. While these initiatives may require coordination across shifts and sites, its long-term impact on capability development and innovation will be significant. The following actions are advised:

- Designated site champions to share successes and challenges across locations
- Offer role shadowing opportunities to allow employees to gain experience in other functions
- Recognize and reward innovations that benefit the network

The expected impact includes reduced loss of institutional knowledge, improved cross-functional understanding, and a stronger culture of collaboration. Feasibility for this intervention ranges from low to high depending on scope and resources, but the benefits still align with Chevalier's BEM and Rummler and Brache's Nine-Variable Model by addressing gaps in knowledge, motives, and process management. By implementing these practices, LMNOP Aviation can create a learning orientated environment that supports adaptability and long-term resilience.

Third Level Recommendations

Third level recommendations address a finer level less impactful for all sites. They address location specific infrastructure needs but will still help increase productivity, capacity, and turnaround time at the affected sites. While the impact of these inventions are limited compared to the systemic changes outlined in our primary and second level recommendations, they remain important for improving operational efficiency at individual locations. These recommendations focus on physical and technological constraints that hinder performance and require targeted investments to resolve.

- Tools/Equipment - ***Resources/Motives***
- Hangar Size/Capacity Restrictions - ***Resources/Organization Goals***

Tools/Equipment

For this third level recommendation it is suggested that LMNOP addresses site-specific tool and equipment issues to eliminate delays and improve operational efficiency. These interventions target physical resource constraints that hinder productivity and turnaround times. The following actions are advised:

- Review and correct tool issues at affected locations
- Provide additional part racks to prevent misplaced components
- Add tool storage sheds to improve organization and accessibility

The expected impact includes fewer work slowdowns, improved workflow organization and reduced risk of errors caused by misplaced or unavailable tools. Feasibility for these actions is generally high, but they do require moderate investment but can be implemented without major changes to the organization. While the scope is limited compared to the systemic recommendations we made above, these improvements are critical for maintaining efficiency and supporting network operations.

Hangar Size/Capacity Restrictions

Finally, our final third level recommendation is evaluating and addressing facility capacity constraints at specific locations to prevent network bottlenecks and lost business. Space limitations currently force some sites to perform work outdoors or redirect aircraft, reducing efficiency and customer satisfaction. The following recommendations are advised:

- Assess hangar layouts and capacity at all locations
- Explore options for reconfiguration, expansion, or relocation at affected sites to increase capacity and eliminate outdoor work

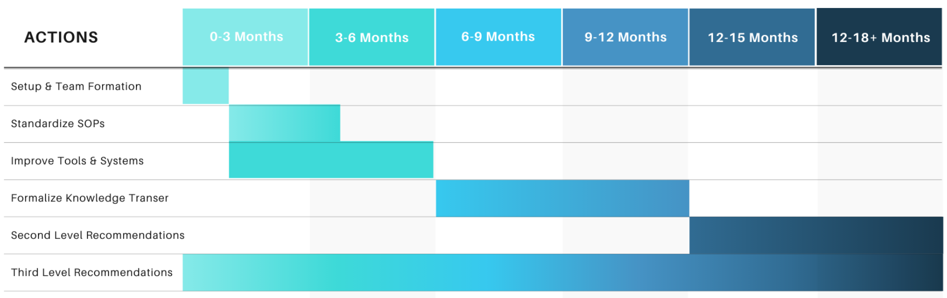
The expected impact includes improved turnaround times, increased capacity, and reduced risk of losing business to competitors with greater infrastructure. Feasibility varies significantly: minor layout adjustments may be relatively low-cost while hangar expansions or relocations represent a high-cost, long-term investment. Despite these challenges, addressing space constraints is essential for sustaining growth and meeting strategic business goals.

Implementation

To implement these interventions, it is essential to have a clear, phased plan that balances ambition with practicality. This roadmap provides a structured approach implementing our primary recommendations: standardizing processes, improving tools and systems, and formalizing knowledge transfer across all sites. The plan is designed for a 12-18 month horizon, allowing time for alignment, adoption, and evaluation while delivering early wins to maintain momentum. Figure 8 provides a visual overview of the plan.

Figure 8

Implementation Gantt Chart



Phase 0: Setup and Team Formation (4-6 weeks)

The first step is to establish a strong foundation for implementing the interventions. A cross-functional working group will be formed, including appropriate representation from groups such as Business Operations, Talent and Development, Service Center Operations Team (Exemplary Locations), system development team, and others who may be able to provide valuable input. This group will define roles and responsibilities and develop standard day-to-day operating procedures to perform tasks associated with the various roles (i.e., TSM, CSM, etc.). To kick off this plan, the team will begin by setting up standard bi-weekly meetings and weekly working sessions to develop these resources and outlines. For this first phase, there will be a few key deliverables that must be completed to move to the next phase. The key deliverable will be living documents designed to be modified as the team progresses.

Key Deliverables:

- Program charter and scope
- Master timeline and milestone plan
- Communication and change management strategy
- Identify key resources that your working group will need

Phase 1: Standardize SOPs (3-4 months)

This next phase focuses on creating consistent, documented processes for critical roles and their associated responsibilities. Some areas already identified from our needs assessment include:

- Project planning, scheduling, and management
- Invoice review and quality control
- Work allocation and assignments

The working group identified above will work together to understand, evaluate, and define best practices focusing on exemplary location practices. To do this, they will conduct process walk-throughs and draft SOPs outlining clear steps, roles, controls, and quality checkpoints. After piloting at select sites, SOPs will be refined and rolled out broadly.

★ *Quick win:* Start with an invoice quality control checklist or job aid and work allocation rules to demonstrate early impact.

Phase 2: Improve Tools and Systems (4-6 months, overlapping Phase 1)

Once SOPs are underway and have a clear path forward, LMNOP's attention should shift to enabling technology that improves the user experience. This includes:

- **Centralize Documentation.** Start with a single, trusted source of information. Some Knowledge Management System (e.g., SharePoint) and create a single-entry web portal containing all SOPs and resources needed to perform particular roles.
- **Develop and Deploy (RAG) LLM Agent.** This will provide quick, accurate answers from trusted, identified sources with links to the knowledge.
- **Assess Existing Tools. T**his would include taking a holistic view of the systems and other resources to limit information in take and make a more user first design.

★ *Quick win:* Find out the biggest pain points in the systems and identify small wins that would take small amounts of time or resources to implement to show improvements.

Phase 3: Formalize Knowledge Transfer (4-6 months starting after Phase 1)

This phase will build capabilities and ensure long-term sustainability for LMNOP’s operation team.

- **Mentoring Program.** Pair experienced staff with new hires for guided learning, either as 30-minute sessions, bridge shifts, or an early retirement program offering new or soon-to-be-retired techs to mentor and support new hires.
- **Role-Specific Onboarding Paths.** Combine SOP training, tool use, and shadowing as a way to develop and train new hires in all roles.
- **Competency Model and Certification Pathway.** Develop a competency model outlining required certifications, skills, and knowledge for each technician grade level, with renewal and performance expectations. Define a pathway for them to achieve these requirements and certifications, so they understand how to progress.

★ *Quick win:* Once requirements have been defined, give TSMs a job aid on how to utilize models for planning and managing a maintenance event based on grade levels.

Phase 4: Second Level Recommendations (12-18 months onward)

Once the primary recommendations are complete, the focus will shift to strengthening leadership, career development, and learning across the network. This will include:

- **Strengthen Performance Management**
  - Train managers on delivering effective, consistent feedback.
  - Establish regular feedback patterns through scheduled 1-on-1 sessions.
  - Utilize recognition programs for excellence.
- **Develop Career Pathways**
  - Adjust leadership compensation and review cost-of-living across locations to remain competitive and make leadership roles more desirable.
  - Define clear competency requirements for advancement and create a leadership development program for aspiring leaders.
  - Offer role shadowing opportunities and enable peer learning through site champions who share successes and challenges.

Phase 5: Third Level Recommendations (Ongoing, location specific)

These recommendations address finer, site-specific infrastructure needs that impact productivity and turnaround time. While less impactful across all sites, they resolve persistent complaints and issues. These recommendations include:

- **Improve WI-FI Connectivity.** Focus on sites with poor network performance.
- **Upgrade Parts Racks, Sheds, and Tools.** This will produce better organization and accessibility.
- **Evaluate Hangar Size and Capacity.** Focus on smaller sites to improve productivity and build a plan around capacity restrictions.

Phase 6: Evaluate and Adjust (Continuous, starts after completion of Phase 1)

After initial implementation to ensure sustained impactful results it is critical to measure impact and adjust as necessary.

Key Actions:

- Conduct quarterly performance reviews of SOP adoption, tools usage, and knowledge transfer effectiveness.
- Use KPIs and dashboards to monitor progress on productivity and invoice accuracy.
  - Compare these results against baseline metrics established during our needs assessment discovery.
- Maintain a continuous improvement backlog for enhancements and quick wins.

Implementation Summary

This phased approach ensures interventions are implemented systemically and with a plan, including early wins to build confidence in the long-term goals and to sustain improvements. This plan is meant to be flexible and adaptive to evolving needs, but is intended to remain a guidepost to remind the team of LMNOP's interventions' focus and goals.

Limitations and Implications

Team Aviation's analysis faced several constraints that affected our result, with potential implications for the reliability and scope of our recommendations.

Remote Data Collection

The team conducted all data collection remotely without on-site observations of work processes. This limitation prevented us from directly observing technician workflows, coordination between TSMs and CSMs, invoice review procedures, and real-time problem-solving approaches. While interviews, surveys, and extant data provided valuable insights, direct observation would have enabled triangulation of findings and a deeper understanding of tacit work practices that participants may not explicitly articulate.

As a result, our recommendations for process standardization and workflow improvements rely primarily on reported practices rather than observed behaviors, which may not fully capture the complexity of actual work execution.

Sampling of Targeted Roles/Sites

Time constraints necessitated a sampling strategy rather than comprehensive assessment of all service centers. We conducted in-depth interviews at a subset of locations representing exemplary and underperforming performance and distributed network-wide surveys to GMs, MMs, CSMs, TSMs, and Technicians. While this approach provided substantial data, we may not have captured location-specific variations, unique challenges at non-interviewed sites, or contextual factors that influence performance differently across the network.

Due to this, recommendations are based on patterns identified across the sample and may require adjustment during implementation to address site-specific conditions not surfaced as part of our analysis.

Potential Response Bias

Time constraints necessitated a sampling strategy rather than a comprehensive assessment of all service centers. We conducted in-depth interviews at a subset of locations representing exemplary and underperforming performance and distributed network-wide surveys to GMs, MMs, CSMs, TSMs, and Technicians. While this approach provided substantial data, we may not have captured location-specific variations, unique challenges at non-interviewed sites, or contextual factors that influence performance differently across the network.

As a result, recommendations are based on patterns identified across the sample and may require adjustments during implementation to address site-specific conditions not surfaced in our analysis.

Limited Review of Support Departments

The project timeline permitted interviews only with TSMs, CSMs, and Technicians. We did not interview support staff, including, parts department personnel, scheduling coordinators, or IT system administrators, all of whom could potentially influence productivity and invoice accuracy. Survey data captured perspectives from General Managers and Maintenance Managers, but a deeper investigation of these roles was not feasible.

As a result, some root causes or intervention opportunities involving cross-functional processes may remain unidentified. Recommendations focus on factors for TSMs, CSMs, and Technicians, but full resolution may require additional analysis of roles we did not assess.

Lack of Competitive Benchmarking Data

Competitor performance data was unavailable, preventing external benchmarking of productivity metrics, invoice accuracy rates, and industry-standard turnaround times. Without market context, we cannot definitively determine whether LMNOP's productivity and invoice accuracy are below, at, or above industry standards. Our original project plan included collecting competitor data, but our research indicates that no such source is available.

As a result, performance targets are based on organizational goals rather than competitive benchmarks. LMNOP may be performing better or worse relative to competitors than internal metrics suggest, affecting the urgency and scale of interventions.

No Access to Compensation Data

Compensation structures, pay scales, and cost-of-living adjustments were not accessible for analysis. Multiple participants reported that leadership compensation is unattractive relative to individual contributor pay, and that cost-of-living variations across locations create equity concerns. However, we could not validate these perceptions against actual pay data or external market benchmarks.

As a result, our recommendation to "adjust leadership compensation and review cost-of-living for locations" is based on employee perceptions rather than a verified compensation analysis. Implementation will require HR to conduct an independent compensation study to determine actual gaps.

Changes to Project Plan

Team Aviation's final needs assessment differed from the original Needs Assessment Planning Table in two key ways:

Original Plan: Conduct interviews at all exemplary and underperforming sites.

- **As Executed:** Selected exemplary and underperforming locations based on data analysis showing clear performance differentiation.
- **Reason:** Data-driven site selection ensured maximum contrast for comparative analysis

Original Plan: Collect competitor benchmarking data.

- **As Executed:** Used organizational goals for benchmarking.
- **Reason:** Industry benchmarking is either unavailable or does not exist.

These adjustments allowed rigor without compromising the fundamental needs assessment approach.

Despite these limitations, Team Aviation has confidence in the needs assessment's findings and recommendations.

Conclusion and Next Steps

This report provides a comprehensive analysis of systemic factors impacting productivity and invoice accuracy across the network. Through triangulated data, root cause analysis, and structured frameworks such as Chevalier's BEM and Rummler & Brache's Nine-Variable, we identified performance gaps and prioritized interventions that address information, resources, and knowledge in the most critical areas revealed by our assessment.

Our recommendations are organized into three tiers:

- Primary Recommendations for immediate, network-wide impact.
- Second-Level Recommendations to strengthen leadership, career development, and organizational learning.
- Third-Level Recommendations for site-specific infrastructure improvements.

A phased implementation roadmap ensures these interventions are executed systematically, balancing ambition with practicality and delivering early wins to maintain momentum. While limitations such as remote data collection and lack of competitive benchmarks may influence scope, the findings provide a strong foundation for measurable improvement.

Next Steps:

- Begin Phase 0 immediately to establish governance and alignment.
- Launch Phase 1 within the next quarter to standardize SOPs and deliver quick wins.
- Monitor progress through KPIs and adjust as needed to sustain impact.

By following this roadmap, the organization can expect significant improvements in productivity, invoice accuracy, and overall resilience, positioning the network for long-term success and sustainability.

References

Rothwell, W. J., Hohne, C. K., & King, S. B. (2018). *Human performance improvement: building practitioner performance*. Routledge.

Rummler-Brache Group (N.D.) *The three levels of performance*. Retrieved from https://www.rummler-brach/default/files.

Stefaniak, J. (2020). *Needs Assessment for Learning and Performance*. Routelage.

Watkins, R., Meiers, M. W., & Visser, Y. L. (2011). *A guide to assessing needs: essential tools for collecting information, making decisions, and achieving development results*.

Appendix A: Codebook and Data

A.1 Codebook Overview

Environment		Information
		Resources
		Incentives
Individual		Knowledge
		Capacity
		Motives
Process		Goals
		Design
		Management
Organizational		Goals
		Design
		Management

A.2 Codebook Definitions

Code	Definition
Clear expectations	Specific, mutually agreed-upon standards, goals, and outcomes
Feedback	Timely information about a person's performance of a task used as a basis for improvement
Communication	Timely, accurate, and accessible flow of information that ensures employees and customers have what they need, when they need it, without bottlenecks or gaps
Job Responsibilities	Related to the specific tasks, duties, and expectations a person is accountable for in their role and/or duties they perform outside their official role
Tools	Access to tools and materials to perform work efficiently
Parts	Related to the procurement and availability of parts
Standard Processes	Related to the presence of clear, well-communicated, standardized processes to carry out work efficiently
Workload	Related to the amount of work assigned to or expected from a person or team within a given period of time.
Environment/Culture/ Relationships	Related to shared norms, values, and working relationships that shape how people treat each other, collaborate, and feel supported in getting work done
Incentives	Financial and non-financial rewards tied to performance measures, that reinforce and encourage desired behaviors and results.
Goal Communication	Related to leaders' awareness of metrics, understanding how metrics align with organizational goals, and how to achieve those goals.
Performance Oversight	Related to leaders actively monitoring results and holding employees to agreed-upon metrics and expectations.
Training/Mentoring	Training or mentoring provided to help employees perform their work efficiently
Skills	Related to knowledge, skills, and experience employees bring to the role or learn on the job.
Operational Efficiency	Related to planning and managing tasks and resources so that work flows smoothly, people are quickly redeployed when conditions change, and goals are met with minimal waste or delay

Work Environment	Related to the physical layout of the site or workspace
Cross-functional Dependency	When work in one team or department can't move forward or be completed until another team or department delivers something it depends on
Attitude	Related to an employee's thoughts and feelings about the work they perform
Capacity	Related to an employee's readiness and fitness to perform, including their ability to learn, their fit with job demands, and the absence of emotional barriers that hinder performance

A.3 Survey Results Overview

	Avg Score	
Environment	3.8	Information
	2.9	Resources
	3.2	Incentives
Individual	3.4	Knowledge
	2.4	Capacity
	3.9	Motives
Process	3.5	Goals
	3.4	Design
	3.3	Management
Organizational	4.9	Goals

	4.2	Design
	4.6	Management

A.4 Organizational Quantitative Data

Organizational quantitative data included, productivity goals vs actuals, Invoice Accuracy goals vs actuals, NOP goals vs actuals, Schedule Adherence, Customer Satisfaction, Hours to standard plan hours, Hours to Planned, and Attrition.

Appendix B: Tool and Instruments

B.1 Frameworks and Models

Chevalier’s BEM - Question Guidebook

Figure 1

	Information	Resources	Incentives
Environmental Factors	- Do employees clearly understand their roles and responsibilities?	- Do employees have access to the tools and matials needed to perform their jobs?	- What type of rewards or recognition are currently in place?
	-Are performance expectations communicated effectively?	- Are there resource shortages that impact performance?	- Do employees feel their efforts are acknowledged and valued?
	- How frequently is feedback provided?	- Is the work environment organized and conducive to productivity?	- Are there opportunities for career growth or job enrichment?
	- Are there guidelines and or documentation available to support task execution?	- Are processes clearly defined and accessible?	- How does the organization support employee motivation?
Individual Factors	Knowledge/Skills	Capacity	Motives
	- Do employees have the necessary training and experience for their roles?	- Do employees feel physically and metally capable of performing their duties?	- Do employees feel personally motivated to perform well?
	- Are there gaps in knowledge or skills that affect performance	- Are job demands reasonable for the employees abilities	- Are individual goals aligned with oragnizational goals?
	- Is ongoing training avaiable and accessible?	- Are accommodations available for employees with specific needs?	- What aspects of the job do employess find most fulfilling or frustrating?
			- Were employees recruited witha clear understanding of the job expectations?

Rummler and Brache’s Nine-Variable Model - Question Guidebook

Figure 2

	Organization Goals	Organization Design	Organization Management
Top Row: Organizational Level	- What are the strategix objectives of the organization?	- Is the organizational structure coducive to achieving strategic goals?	- How does leadership monitor organizational performance?
	- How are these goals communicated across departments?	- Are departements and teams aligned and coordinated effectively?	- What systems are in place for decision-making and governance?
	- Are performance metrics aligned with strategic goals?	- Are there clear lines of authority and accountability?	- How are resources allocated across the organization?
	- How does leadership prioritize these goals?	- How flexible is the organization in adapting to change?	How does leadership respond to performance issues?
Middle Row: Process Level	Process Goals	Process Design	Process Management
	- What are the expected outcomes of this process?	- Is the process clearly defined and documented?	- Who is responsible for managing this process?
	- How are process goals communicated to employees?	- Are roles and responsibilities within the process well understood?	How is performance monitored and reported?
	- Are there metrics in place to measure process success?	- Are there bottlenecks or inefficiencies in the workflow?	- Are there feedback loops to improce the process?
	- How are the process goals aligned with organization goals?	How often is the process reviewed or updated?	- What happens when the process fails or underperforms?

B.2 Analysis Tools

The Five Whys Analysis

Productivity

Problem Statement: Less than half of the centers are meeting their individual productivity goal set by the organization.				
Why #	**Why?**	**Answer**	**Evidence**	**Categories**
1	Why are only some centers meeting their productivity goals?	Many centers have a growing percentage of new and less experienced technicians who take longer and need more support to complete work.	Interviews, extant data	• Individual - Knowledge/Skills; Individual - Capacity • Performer Design (skill requirements vs

				actual experience mix)
2	Why do new and less experienced technicians stay slower and more dependent instead of ramping up quickly?	New technicians have fewer learning opportunities due to scheduling norms, so they are not consistently paired with experienced techs or given time for structured coaching.	Interviews	• Environment - Resources (work design, scheduling), plus a secondary impact on Individual - Knowledge/Skills • Process Design (how work and mentoring opportunities are structured)
3	Why do scheduling norms limit learning opportunities for new technicians?	Training and development practices are inconsistent across centers, so each site handles mentoring and skill development differently.	Interviews; survey data	• Environment - Resources (training systems, methods) • Organization Design (overall learning and development system)
4	Why is training inconsistent across centers?	There is a lack of formal knowledge transfer systems that define standard training content, methods, and responsibilities for sharing expertise.	Interviews; survey data	• Environment - Resources (knowledge transfer mechanisms, SOPs) • Process Design; Organization Management (no standard

				process or clear ownership for knowledge transfer)

Problem Statement: Less than half of the centers are meeting their individual productivity goal set by the organization.				
Why #	**Why?**	**Answer**	**Evidence**	**Category**
1	Why are only some centers meeting their productivity goals?	Technicians spend extra non-billable time searching for manuals and job information because it is spread across multiple portals and systems.	Interviews; System Review	• Environment - Resources; Environment - Information • Process Design; Performer Design
2	Why do technicians need to search across multiple portals and systems for the information they need?	There is no single, consolidated source of truth or standard access path for technical information, so techs must log into multiple OEM sites and internal systems	Interviews; System Review	• Environment - Resources; Environment - Information • Process Design
Note: Due to time and access constraints, the team did not investigate deeper governance or ownership causes behind information fragmentation. For the purposes of this project, “manuals/information spread across multiple portals” is treated as a practically actionable root cause and a direct target for intervention design.				

Problem Statement: Less than half of the centers are meeting their individual productivity goal set by the organization.				
Why #	**Why?**	**Answer**	**Evidence**	**Category**

1	Why are only some centers meeting their productivity goals?	Processes and maintenance project management practices vary significantly by site, so work is planned, scheduled, and tracked differently and productivity is inconsistent.	Interviews; survey comments	• Environment - Resources (processes, SOPs); Environment - Information (how work is tracked) • Process Design; secondary Performer Design (for managers)
2	Why do processes and maintenance project management practices vary so much by site?	Processes and maintenance project management (organizational SOPs) are not standardized across sites, so managers must decide on their own how to proceed in most situations.	Interviews; survey comments	• Environment - Resources (standard procedures); Environment - Information (guidance) • Organization Design (lack of standard system); Process Design
Note: Due to time and access constraints, the team did not investigate deeper governance or ownership causes behind the lack of standardized processes. For the purposes of this project, “Processes and maintenance project management (organizational SOPs) are not standardized across sites” is treated as a practically actionable root cause and a direct target for intervention design.				

Problem Statement: Less than half of the centers are meeting their individual productivity goal set by the organization.				
Why #	**Why?**	**Answer**	**Evidence**	**Category**

1	Why are only some centers meeting their productivity goals?	TSMs struggle to assign the right technicians to the right work. This can cause delays if scheduled technicians don't have the necessary skills for the project. Centers that perform better tend to have TSMs who know their technicians' skills well and can assign work more accurately.	Interviews; survey data/comments	• Environment - Resources (assignment/scheduling practices); Environment - Information (who can do what) • Process Design (how work is allocated); Performer Design (TSM role)
2	Why are jobs more often assigned to technicians without the needed skills at some centers?	At those centers, TSMs do not have a clear, up-to-date picture of each technician's specific skills and experience, so they rely on guesswork or incomplete information when assigning work.	Interviews; survey data	• Environment - Information (visibility of individual skills); secondary: Environment - Resources (no simple way to see skills)
3	Why don't TSMs have a clear, up-to-date picture of technicians' skills and experience?	There is no consistent, organization-wide way to document and maintain technician competencies and task capabilities (e.g., skills matrix, competency standards by grade level), so knowledge is mostly informal	Interviews; extant data	• Environment - Resources (no skills matrix/system); Environment - Information (no formalized skill definitions or records) • Organization Design (role structure and grading);

		and varies by site and manager.		Process Management (how skills are tracked)
4	Why has a standardized skills matrix / competency model not been implemented across centers?	Grade levels and work assignments have historically been based on tenure, certifications, or local manager judgment rather than standardized observable competencies.	Interviews; Extant Data	• Environment - Information (unclear role definitions); Environment - Resources (no process for updating) • Organization Management; Organization Design

Problem Statement: Less than half of the centers are meeting their individual productivity goal set by the organization.				
Why #	**Why?**	**Answer**	**Evidence**	**Category**
1	Why are only some centers meeting their productivity goals?	Employees at several centers spend extra time navigating, troubleshooting, and updating computer tools for everyday tasks, reducing available time for billable work, technician oversight, managing projects, etc. and sometimes lose time when unexpected changes appear in the system.	Interviews; survey data/comments	• Environment - Resources (software, workflow tools); Environment - Information (how work is captured) • Process Design (workflows through tools); secondary Performer

				Design (role expectations)
2	Why do technicians and managers spend extra time working in the computer tools instead of completing billable/management work?	Sites are becoming "paper free" and doing all work through computer software, but these tools are not user-friendly, require frequent updates and time-consuming manual processes. Updates are sometimes pushed with little notice or inconsistent communication, so some employees only discover changes when they log in. Proficiency with the tools also varies across sites.	Interviews; survey data/comments	• Environment - Resources (tool usability, workflow fit); Environment - Information (change notices, guidance); Individual - Knowledge & Skills (tool proficiency) • Process Design; Performer Support (training & comms about tools)
Note: Due to time and access constraints, the team did not investigate deeper design, selection, or change-management causes behind the current computer tools and "paper free" workflows. For the purposes of this project, "Sites are becoming paper free and doing all work through computer software, but these tools are not user-friendly, require frequent updates and time-consuming manual processes, and updates are sometimes pushed with little notice or inconsistent communication, leading to varying levels of proficiency and unexpected changes" is treated as a practically actionable root cause and a direct target for intervention design.				

Problem Statement: Less than half of the centers are meeting their individual productivity goal set by the organization.				
Why #	**Why?**	**Answer**	**Evidence**	**Category**
1	Why are only some centers meeting their productivity goals?	Technicians lose time looking for, waiting for, or	Interviews	• Environment - Resources (tools, physical

		improvising around physical tools that are outdated, not readily available, or in poor condition (being used by another team, moved across the hangar, etc).		equipment); secondary Environment - Information (where tools are at) ● Process Design; Performer Design
2	Why do technicians encounter outdated, poorly maintained, or missing/unavailable tools?	There are not enough of the right tools at the point of use, tools are not consistently maintained or replaced	Interviews; survey data/comments	● Environment - Resources (inventory, maintenance); Environment - Information (no simple way to know status/location) ● Process Design (tool provisioning); Organization Design (how tools are managed)
Note: Due to time and access constraints, the team did not investigate deeper governance, budgeting, or ownership causes behind physical tool availability and condition. For the purposes of this project, "There are not enough of the right tools at the point of use, tools are not consistently maintained or replaced, tool sharing across teams/hangar is managed ad hoc without clear rules or tracking" is treated as a practically actionable root cause and a direct target for intervention design.				

Problem Statement: Less than half of the centers are meeting their individual productivity goal set by the organization.				
Why #	**Why?**	**Answer**	**Evidence**	**Category**

1	Why are only some centers meeting their productivity goals?	At several centers, work is slowed when aircraft cannot be moved in and out efficiently; teams lose time reshuffling aircraft, waiting for space to open up, or redirecting aircraft to other locations when the hangar reaches capacity.	Interviews; survey data/comments; extant data on hangar sizes by location	• Environment - Resources (physical workspace, bays); secondary: Environment - Information (status of bays/slots) • Performer Environment (local conditions); Process Design (flow of work through hangar)
2	Why do aircraft frequently need to be reshuffled or redirected, creating delays and capacity issues?	Hangar design constraints and limited space create release delays and capacity issues. Sites must reshuffle aircraft to extract completed planes or redirect incoming aircraft to other locations when the hangar reaches capacity.	Interviews; survey data/comments; extant data on hangar sizes by location	• Environment - Resources (facility size/layout, number of bays); Environment - Information (no simple way to visualize/manage constraints) • Performer Environment; Organization Design (physical infrastructure)
Note: Due to time and access constraints, the team does not have data on capital planning and network design. We stop at “hangar constraints and limited space” and treat that as a practically actionable root cause for this project.				

Problem Statement: Less than half of the centers are meeting their individual productivity goal set by the organization.				
Why #	**Why?**	**Answer**	**Evidence**	**Category**
1	Why are only some centers meeting their productivity goals?	At some centers, technicians repeat the same inefficiencies and errors because the feedback and coaching they receive from managers is vague, inconsistent, or infrequent, which limits their ability to improve productivity.	Survey data; survey comments	● Environment - Information (quality of feedback); secondary: Individual - Knowledge/Skills (techs don't know what to change) ● Performer Management (how tech performance is coached); Process Management (how performance is monitored)
2	Why do technicians receive such different quality of feedback and coaching across centers?	Manager feedback quality varies significantly depending on the individual manager; some TSMs give specific, timely, behavior-based feedback, while others provide mostly reactive or general comments, so technician development depends heavily on who their manager is.	Survey data; survey comments	● Environment - Information (manager practices); Individual - Knowledge/Skills (coaching skill of managers) ● Performer Management (TSM coaching behavior); Organization Management (variation by leader)

3	Why does feedback and coaching style depend so heavily on the individual manager?	There are no clear, organization-wide expectations, standards, or training for how TSMs should provide performance feedback and coaching to technicians (e.g., frequency, content, use of data), so each manager relies on their personal style and experience.		• Environment - Information (lack of expectations/standards); Individual - Knowledge/Skills (varies with individual manager) • Performer Design; Organization Management
4	Why are there no clear expectations or training for TSMs on giving feedback?	The TSM role is mainly structured around managing work, scheduling, and output metrics. Coaching/feedback is treated as a “nice to have” rather than a core responsibility with tools, support, and time to complete.		• Environment - Information; Environment - Resources • Performer Design; Organization Design

Invoice Accuracy

Problem Statement: Half of the sites are not meeting their invoice accuracy goal set by the organization.				
Why #	**Why?**	**Answer**	**Evidence**	**Category**
1	Why are only half the centers meeting the	Invoices are sent with preventable errors (coding,	Extant data; interviews; survey data	• Environment - Information (quality of

	invoice accuracy goal?	billable hours, parts, rates, etc.)		information on invoice); • Process Design (how invoices are produced); Performer Management (how work is checked)
2	Why are invoices sent with preventable errors?	No standard processes for invoice review and quality control. Each site or individual handles it differently and some invoices receive little or no systematic check before being sent to customer	Interviews, survey data/comments	• Environment - Resources (no standard process) • Process Design; Process Management
Note: Due to time and access constraints, the team did not investigate deeper governance, ownership, or cross-functional causes behind absence of standard invoice review processes. For the purposes of this project, “There are no standard processes for invoice review and quality control; each site or individual handles review differently, and some invoices receive little or no systematic check before being issued” is treated as a practically actionable root cause and direct target for intervention design.				

Problem Statement: Half of the sites are not meeting their invoice accuracy goal set by the organization.				
Why #	**Why?**	**Answer**	**Evidence**	**Category**

1	Why are only half the centers meeting the invoice accuracy goals?	Invoices are sent with preventable errors (coding, billable hours, parts, rates, etc.)	Extant data; interviews; survey data	● Environment - Information (quality of information on invoice); ● Process Design (how invoices are produced); Performer Management (how work is checked)
2	Why are preventable errors not consistently identified and corrected before invoices are sent?	There is a lack of documentation and practical guidelines for invoice review, so CSMs do not have clear, written criteria or checklists to follow and often rely on personal judgment or habits.	Interviews; survey data	● Environment - Information (missing standards/guidelines); Environment - Resources (no checklists/templates) ● Process Design (no defined review artifacts); Performer Expectations (unclear “what to check”)
Note: Due to time and access constraints, the team did not investigate deeper governance, ownership, or cross-functional design causes behind the lack of documented invoice review guidelines. For the purposes of this project, “There is a lack of documentation and practical guidelines for invoice review, so reviewers do not have clear, written criteria or checklists to follow and often rely on personal judgment or habits” is treated as a practically actionable root cause and a direct target for intervention design.				

Why #	Why?	Answer	Evidence	Category
	Problem Statement: Half of the sites are not meeting their invoice accuracy goal set by the organization.			
1	Why are only half the centers meeting the invoice accuracy goals?	Staff at some sites make recurring invoice mistakes (e.g., hours, parts, rates, coding) because they do not fully understand invoice requirements, system behaviors, or common error traps.	Extant data; interviews; survey data	• Individual - Knowledge/Skills (how to invoice correctly); Environment - Information (errors only caught downstream) •
2	Why don’t staff fully understand invoice requirements and common error traps?	Many employees learn invoice tasks through trial-and-error, with no or inadequate role-specific training on how to create accurate invoices in the systems used.	Interviews; survey data/comments	• Individual - Knowledge/Skills (lack of targeted training); Environment - Resources (no structured training content/job aids) •
3	Why is role-specific training on invoice accuracy absent or inadequate?	There is no standardized onboarding or training curriculum for CSMs	Interviews; survey data	• Environment - Resources (no standard curriculum/materials); Environment - Information (no defined learning objectives) •
	Due to time and access constraints, the team did not investigate deeper governance, ownership, or cultural causes behind the lack of standardized onboarding and training for			

CSM roles. For the purposes of this project, "There is no standardized onboarding or training curriculum for CSM roles" is treated as a practically actionable root cause and a direct target for intervention design.

Problem Statement: Half of the sites are not meeting their invoice accuracy goal set by the organization.				
Why #	**Why?**	**Answer**	**Evidence**	**Category**
1	Why are only half the centers meeting the invoice accuracy goals?	At some sites, invoices are sent with preventable errors because they are not thoroughly reviewed; CSMs are rushing or skipping checks to keep work moving.	Interviews; survey data	● Environment - Information (errors not caught pre-send) ● Process Design (no robust QC step in practice); Performer Management (how work is checked)
2	Why are CSMs rushing or skipping invoice checks?	At some sites, TSMs are also performing CSM/invoicing responsibilities, so they are overworked and must balance scheduling, customer communication, and invoicing. They do not have enough time to thoroughly review each invoice.	Interviews; survey data	● Individual - Capacity (time/attention vs workload); Environment - Resources (time pressure to "get invoices out") ● Performer Capacity; Performer Design (role overload)

Note: For this analysis, the combination of overworked TSMs also performing CSM/invoicing responsibilities, leaving insufficient time for thorough invoice review, is treated as a practically actionable root cause and a direct target for intervention design.

Problem Statement: Half of the sites are not meeting their invoice accuracy goal set by the organization.

Why #	Why?	Answer	Evidence	Category
1	Why are only half the centers meeting the invoice accuracy goals?	Complex invoices for maintenance work include preventable technical errors (labor, parts, task coding) that are not caught before billing, especially on higher complexity jobs.	Extant data; interviews; survey data	● Individual - Knowledge/Skills (understanding of work content); Environment - Information (complex information to interpret) ● Performer Knowledge and Skills; Process Design (how some technical work is translated into codes)
2	Why are technical errors on complex invoices not caught by the CSM?	Many CSMs lack the technical aviation knowledge needed to understand complex maintenance tasks, parts, and configurations, so they cannot reliably verify whether the	Interviews; survey data	● Individual - Knowledge/Skills (technical understanding); Environment - Resources and Tools (limited technical/training or reference aids)

		invoice content makes sense.		• Performer Design; Performer Support
3	Why do many CSMs lack sufficient technical aviation knowledge?	CSMs are hired and developed primarily for customer service, coordination, and administrative skills, and receive limited structured training on aircraft systems, maintenance concepts, and how those map to invoice line items and billing codes.	Interviews; survey data; extant data	• Individual - Knowledge/Skills; Environment - Resources (no structured curriculum) • Organization Design (role profile for CSM); Performer Support (training and development)
Due to time and access constraints, the team did not investigate deeper governance, role design, or cultural causes behind how CSM capabilities are defined and developed. For the purposes of this project, “CSMs are hired and developed primarily for customer service, coordination, and administrative skills, and receive limited structured training on aircraft systems, maintenance concepts, and how those map to invoice line items” is treated as a practically actionable root cause and a direct target for intervention design.				

Problem Statement: Half of the sites are not meeting their invoice accuracy goal set by the organization.				
Why #	**Why?**	**Answer**	**Evidence**	**Category**
1	Why are only half the centers meeting the invoice accuracy goals?	Staff make preventable data-entry and coding errors (e.g., wrong task codes, incorrect labor/parts lines, missed fields) when	Extant data; interviews; survey data	• Environment - Data (errors in what’s entered); secondary: Individual - Knowledge/Skil

		creating or editing invoices in the system.		ls (how to navigate) ● Process Design (how invoices are entered/edited in tools); Performer Management (how work is checked)
2	Why are staff making preventable data-entry and coding errors in the system?	The computer tools used for invoicing are not user-friendly: technical reason codes are hard to interpret, navigation is clunky, and it is easy to click the wrong option or miss required information when under time pressure.	Interviews; survey data/comments	● Environment - Resources (system usability, screen design); Environment - Information (how information is presented) ● Process Design;
Note: Due to time constraints, the team did not go deeper into governance/ownership of tool design. For the purposes of this project, "The computer tools used for invoicing are not user-friendly" is treated as a practically actionable root cause and a direct target for intervention design.				

Problem Statement: Half of the sites are not meeting their invoice accuracy goal set by the organization.				
Why #	**Why?**	**Answer**	**Evidence**	**Category**
1	Why are only half the centers meeting the invoice accuracy goals?	Invoice errors persist over time; the same types of mistakes (hours, parts, coding) reoccur	Interviews; survey data; extant data	● Environment – Information (how errors are addressed; error patterns

		without consistent behavior change at the individual or site level.		not driving change) • Performer Design; Process Management (how performance is monitored)
2	Why do the same invoice errors and rework patterns keep recurring?	There is a weak accountability culture around invoice accuracy: expectations are not consistently reinforced, errors are often treated as routine "cleanup," and individuals or teams rarely experience clear follow-up, coaching, or consequences when error rates are high.	Interviews; survey data	• Environment - Information (limited follow-up, low perceived consequences); Individual - Motives (low perceived importance of accuracy) • Performer Management; Organization Management (how leaders respond)
Note: Due to time and access constraints, the team did not investigate deeper governance, performance management, or cultural causes behind recurring invoice errors. For the purposes of this project, "There is a weak accountability culture around invoice accuracy: expectations are not consistently reinforced, errors are often treated as routine cleanup, and individuals or teams rarely experience clear follow-up, coaching, or consequences when error rates are high" is treated as a practically actionable root cause and a direct target for intervention design.				

Problem Statement: Half of the sites are not meeting their invoice accuracy goal set by the organization.

Why #	Why?	Answer	Evidence	Category
1	Why are only half the centers meeting the invoice accuracy goals?	Invoices are released with preventable data and coding errors (e.g., incorrect codes, mismatched labor/parts, missing fields) that are only discovered after billing.	Environment - Data (error-prone entries); Environment - Information (errors caught late)	● Environment - Data (error-prone entries); Environment - Feedback & Consequences (errors caught late) ● Process Design (how invoices are created/released); Performer Management (how work is checked)
2	Why are these errors not consistently caught before invoices are released?	The system design of coding and billing tools allows errors to pass through uncaught rather than preventing them proactively; many incorrect or incomplete combinations are technically allowed, and invoices can be finalized without robust validation.		● Environment - Resources (system behavior, validation rules); Environment - Information (what the system enforces) ● Process Design (workflow and controls in the tools); Performer Environment (system as part of local conditions)

Note: Due to time and access constraints, including limited access to IT and system decision makers, the team did not investigate deeper governance or configuration causes behind the current design of coding and billing tools. For the purposes of this project, "The system design of coding and billing tools allows errors to pass through uncaught rather than preventing them proactively" is treated as a practically actionable root cause and a direct target for intervention design.

Problem Statement: Half of the sites are not meeting their invoice accuracy goal set by the organization.

Why #	Why?	Answer	Evidence	Category
1	Why are only half the centers meeting the invoice accuracy goals?	CSMs often do not have a complete, clear picture of what was actually done on a maintenance job, so it's difficult for CSMs to get accurate details about work performed.	Interviews	• Environment - Information (incomplete/unclear information); secondary: Individual - Knowledge/Skills (can't reconstruct the work from records) • Process Design (how work info flows into invoicing); Performer Management (how work is summarized)
2	Why don't CSMs have a complete, clear picture of what was done on a maintenance job?	Communication and information about a single project are spread across multiple tools and channels with inconsistent access	Interviews	• Environment - Information (fragmented, inconsistently captured info); Environment - Resources

		across stakeholders. Sometimes updates come via email or text, information from phone calls is not consistently recorded, details are often vague, and there is no set communication process or one place where all information is aggregated.		(multiple tools, no single source of truth) • Process Design (information capture and handoff)
3	Why is project communication so fragmented across tools and channels with no single aggregated record?	There is no standardized cross-functional communication workflow or designated “system of record” for maintenance project information, so teams use whatever channels are most convenient (email, text, calls, ad hoc notes) and rely on individuals to piece together the story.		• Environment - Resources (no defined primary tool/workflow); Environment - Information (no standard for what must be captured) • Organization Design (missing process definition); Process Design (how information should flow)

Note: Due to time and access constraints, the team did not investigate deeper governance, ownership, or cultural causes behind how cross-functional project communication is designed and managed. For the purposes of this project, “There is no standardized cross-functional communication workflow or designated ‘system of record’ for maintenance project information, so teams use whatever channels are most convenient (email, text, phone, ad hoc notes) and rely on individuals to piece together the story” is treated as a practically actionable root cause and a direct target for intervention design.

SWOT Analysis

Theme 1: Process and SOP Standardization

- **Core observation**: Inconsistent processes are preventing the network from fully converting paid labor into reliable, billable output and accurate invoices, despite having strong local examples of what “good” looks like.
- **Strengths:**
    - Local autonomy allows high-performing centers to innovate and refine efficient processes
    - Existing “exemplary centers” provide proven practices that can be modeled and scaled
    - Frontline teams have deep, practical knowledge of how to plan work, estimate hours, and sequence activities effectively in their specific context
    - CSMs and invoice staff have developed peer review habits (checking with more experienced colleagues or the Invoice Review Team) when they are unsure, which shows a willingness to seek quality.
    - There is demonstrated appetite for structure: some participants expressed a desire for clearer processes, checklists, and decision support for both maintenance planning and invoicing
    - Existing systems (e.g., maintenance planning tools, invoice systems) can act as delivery channels for embedding standard steps and checks once those are defined
- **Weaknesses:**
    - Variable processes drive inconsistent labor utilization, billable hours, and turnaround times
    - Harder to benchmark centers and forecast capacity
    - Increased risk of rework, non-billable time, and uneven quality across the network
    - Invoice review processes are highly variable; some invoices receive minimal checks, and many CSMs lack access to clear checklists and job aids, increasing risk of preventable errors
    - Process knowledge is person-dependant: when local leaders or experienced CSMs change roles, processes change with them, undermining stability and predictability
    - Because SOPs are fragmented or absent, it is difficult to measure adherence and distinguish between performance issues and process design issues
    - Difficult to create standardized, scalable training and clear performance expectations
- **Opportunities:**
    - Industry best practices, audit frameworks, and process templates can be adapted to accelerate SOP design for maintenance and invoicing

    - Corporate-level process improvement, quality, or continuous improvement teams outside the service group can provide facilitation, methodology, and support.
    - Vendor and system partners (e.g., ERP, system, billing) often offer standard workflows and configuration options that can be leveraged as a starting point instead of starting from scratch.
- **Threats:**
    - Lost revenue and margin if inconsistent processes continue to impact productivity
    - Customer dissatisfaction due to unpredictable turnaround times and variable quality
    - Increased risk of safety, compliance, or warranty issues from inconsistent practices
    - Competitors with standardized, efficient operations may gain a cost and service advantage
- **Impact:** Without standardizing and scaling what works at high-performing sites, the organization risks leaving revenue on the table, eroding customer trust, and falling behind competitors with more consistent, efficient operations.
- **Levers:**
    - Capture and standardize best practices from top-performing centers into a common operating model
    - Implement aligned digital tools, SOPs, and training that support consistent execution

Theme 2: Information and knowledge/communication management

- **Core observation**: Fragmented communication increases delays, rework, and downtime for maintenance and leads to less accurate billing.
- **Strengths:**
    - The information generally exists somewhere (manuals, portals, texts, Teams), so gaps are often about finding and aligning information rather than a total absence
    - Some roles report basic confidence in documentation
    - Teams have created local workarounds (spreadsheets, informal handoffs, checking every work order) that show problem awareness and willingness to compensate
    - Existing digital tools provide foundation to build on, rather than starting from zero
- **Weaknesses:**
    - No central system of record for maintenance events; information is spread across multiple computer programs, plus texts, emails, phone calls, Teams, and sticky notes
    - Cross-functional and cross-shift communication is inconsistent; offshift updates and changes to maintenance discrepancies or departure dates are often missed or not in shared tools
    - Staff must manually reconcile multiple sources to understand what was done, which consumes time and introduces human error.

- New processes and changes are perceived as poorly communicated or late, leading to confusion and uneven adoption across sites.
- Because communication practices vary by site and even by individual, visibility into work in progress and status is limited, complicating scheduling, planning, and invoicing.

- **Opportunities:**
  - Growing customer expectations for transparent, traceable communication (status updates, clear documentation, digital records) create a compelling business case for improving information management.
  - Industry best practices for shift handoffs, MRO documentation, and customer communication can be adapted rather than invented from scratch.
- **Threats:**
  - Competitors that offer better visibility and communication (e.g., real-time status, consistent updates, clear documentation) may be perceived as more reliable and easier to work with.
  - Errors caused by communication and lack of documentation can lead to external audits
  - Employees may move to competitors who utilize more structured communication and digital systems
  - As volume and complexity grow, the current fragmented approach may become unsustainable at scale, leading to more frequent misses, delays, and billing disputes.
  - Staff turnover can result in loss of “who knows where things live” knowledge, making an already fragmented system even harder to navigate.
- **Impact:** Fragmented information and ad hoc communication practices force staff to waste time searching and re-confirming details, create blind spots in planning and handoffs, and increase the risk of missed or misapplied updates, directly reducing productivity and contributing to invoice errors, rework, and customer frustration.
- **Levers:**
  - Define and implement a single system of record for each maintenance event (even if underlying systems remain separate), with clear rules about what must be captured there.
  - Design a standard cross-functional communication workflow (who records what, where, and when) for maintenance discrepancies, schedule changes, service coverage decisions, and handoffs between shifts.
  - Create simple, consistent shift-handoff templates and embed them in existing tools (e.g., within the primary work-order or scheduling system).
  - Use automation where possible (e.g., alerts or flags when maintenance discrepancies are added or departure dates change) to reduce reliance on manual checking.
  - Provide targeted training and job aids on “where to put what” and how to use the designated system of record, reinforcing that email/text are supplements, not the system.

Theme 3: Digital Tools and System Design

- **Core observation**: The shift to “paper free” operations has increased reliance on digital tools for planning, executing, and billing maintenance work, but many of these tools are not user-friendly, require time-consuming manual steps, and are configured with limited validation. Technicians and CSMs must navigate multiple screens, confusing fields, and frequent, poorly communicated updates, which slows daily work and makes it easy for incomplete or incorrect entries to pass through to the invoice.
- **Strengths:**
    - The organization has already invested in core digital platforms for maintenance, scheduling, and invoicing, so the foundation exists; the issue is configuration and usability, not starting from scratch.
    - Staff have shown a high degree of adaptability and resourcefulness in learning and using the tools despite friction, including developing local tricks/workarounds and helping each other navigate complex screens.
    - Moving away from paper creates potential for better traceability, searchability, and reporting once workflows and interfaces are improved.
    - Some system capabilities (e.g., flags, required fields, standard codes) can be reconfigured to support standard processes and validation for both productivity and invoice accuracy.
- **Weaknesses:**
    - User interfaces are often clunky and unintuitive (multiple screens, confusing labels, non-linear workflows), increasing time per transaction and cognitive load.
    - Many workflows depend on manual data entry and re-entry, with limited use of automation, templates, or defaults, which increases the risk of errors and slows throughput.
    - System design allows incorrect or incomplete combinations (e.g., mismatched codes, missing data) to be saved and even finalized, rather than preventing errors proactively.
    - Updates and changes to tools are pushed with little notice or explanation, and proficiency with tools varies widely across sites, which leads to inconsistent usage and frustration.
    - Digital tools are not consistently aligned with standardized processes (because those are still emerging), so the system reinforces variation rather than reducing it.
- **Opportunities:**
    - Industry wide modernization of systems allows the adoption of integrating better performing platforms in the aviation industry.
    - Emerging tools (workflow automation, guided tools, in-app help) can help reduce manual entry and help less-experienced users complete complex tasks accurately.
    - Vendors and platform providers often offer usability improvements, configuration options, and validation rule frameworks that can be leveraged to improve screens and workflows.

    - Corporate IT, digital transformation, or analytics teams outside the service group may be able to support UX redesign, integration, automation, and reporting, reducing load on internal IT teams.
- **Threats:**
    - Competitors with more mature, user-friendly systems may be able to process work faster, with fewer errors, improving both turnaround times and billing accuracy.
    - Vendor limitations, slow development cycles, or costly upgrades could constrain how quickly LMNOP can improve usability and validation.
    - Increasing complexity of aircraft, programs, and billing rules may outpace current tool configuration, making manual workarounds more common and more fragile.
    - As reliance on digital systems deepens, system outages, performance issues, or integration failures have greater potential to disrupt operations and invoicing.
- **Impact:** Non-user-friendly, non-optimized digital tools add friction to almost every step of the workflow: technicians and TSMs lose productive time navigating and re-entering data, while CSMs and invoice staff must work harder to assemble and verify information. This slows maintenance throughput, increases error rates in coding and billing, and drives rework and customer frustration when invoices do not align with the work performed.
- **Levers:**
    - Conduct user-centered reviews of key maintenance and invoicing workflows to simplify navigation, reduce clicks, and align screens with real work experience.
    - Implement stronger validation rules and business logic (e.g., required fields, allowed combinations, automated checks) to prevent common errors from reaching the invoice.
    - Embed standard process steps into the tools (status, checklists, prompts), so the system actively supports the desired way of working.
    - Improve change management for system updates (advance notice, quick-reference guides, microlearning), to reduce disruption and variability in usage.
    - Develop super-user or champion roles at each site to support peers, surface usability issues, and partner with IT on continuous improvement.

Theme 4: Training, onboarding, and capability systems for key roles

- **Core observation**: Capability for both technicians and CSMs is built largely through informal, locally-driven learning rather than consistent, organization-wide training and competency systems. This creates wide variation in skills across sites and shifts, slows ramp-up, and increases reliance on a shrinking pool of experienced staff to keep maintenance moving and invoices accurate.
- **Strengths:**
    - Many technicians feel basically able to do their current jobs and know where to go for help, reflecting strong informal support networks and willingness to coach from more experienced staff.

  - Some sites already practice effective mentoring and pairing of new technicians with experienced ones, providing a model to build on.
  - There are existing resources for product and systems training (OEM materials, internal courses), even if not consistently deployed.
  - On the CSM side, many veterans bring strong customer and coordination skills, and some have TSM or technical backgrounds that could serve as subject matter experts for more formal training.
  - Both groups show recognition of gaps and desire for more structure (clearer grade expectations, invoice training,technical reason code guidance).
- **Weaknesses:**
  - No standardized, organization-wide curricula for technicians or CSMs, on-the-job training quality and content varies significantly by site and manager.
  - At some sites, new technicians are often scheduled on off-shifts with limited senior coverage, weakening real-time coaching and leading to work stoppages when complex issues arise.
  - Cross-training on aircraft is inconsistent, and manual quality varies, so newer technicians depend heavily on local tribal knowledge, which is uneven and vulnerable to turnover.
  - CSMs report minimal formal training for complex tasks, and rely on peers and “trial by fire,” contributing to invoice errors and rework.
  - Capability systems (skills matrices, progression paths, structured knowledge transfer), are fragmented or missing, making it difficult to plan development, accurately forecast work capacity, and scale good practices.
- **Opportunities:**
  - Military and veteran pipelines, apprenticeship models, industry associations, and partnership with aviation schools can help attract and develop new technicians more systematically.
  - External competency frameworks and training standards (for maintenance, customer-facing roles, and billing), can be adapted rather than built from scratch
  - Digital learning tools (microlearning, simulations, guided practice in systems) can support scalable, consistent training across sites and shifts.
- **Threats:**
  - Chronic shortages of A&P technicians, aging workforce, and difficulty attracting new talent limit available maintenance capacity and make it harder to meet productivity expectations.
  - Rising labor costs as competition increases for technicians due to labor shortage.
  - Increased risk that as LMNOP trains new people, competitors will recruit them away.
  - Growing complexity of aircraft, avionics, and warranty rules raises the bar for both technical and billing knowledge, making informal learning increasingly insufficient.
  - Customer expectations for credible technical advice and clean, accurate invoices may drive them to competitors if CSMs and technicians are perceived as underprepared.

  - Regulatory and OEM scrutiny may tighten expectations for documented qualifications and training, making ad hoc learning riskier over time.
- **Impact:** Inconsistent training, tenure-based grade leveling, and weak knowledge-transfer systems slow ramp-up, create wide variation in capability across sites and shifts, and concentrate critical know-how in a shrinking group of experienced staff. This reduces productivity (through slower problem-solving, work stoppages, and uneven use of capacity) and undermines invoice accuracy (through coding errors, program-coverage mistakes, and extra handoffs back to TSMs and leads).
- **Levers:**
  - Define competency models and grade-level skills expectations for technicians and CSMs, and use them to guide assignments, feedback, and development.
  - Build standardized onboarding and training curricula for key roles (techs and CSMs), including structured practice on real scenarios and use of systems.
  - Implement skills matrices at site and network level to give TSMs clear visibility into who can do what and to plan cross-training deliberately.
  - Formalize and scale mentoring/knowledge-transfer practices (e.g., structured pairing, off-shift coverage by experienced techs, communities of practice).
  - Capture and leverage experienced technicians' knowledge into reusable artifacts (playbooks, checklists, videos, SME sessions).
  - Develop focused job aids and reference tools (fault-code guides, service coverage decision trees, invoice review checklists) to reduce reliance on tribal knowledge.

Theme 5: Performance management and culture

- **Core observation**: Management quality, expectations, oversight and leadership vary across sites and shifts. Some employees report supportive leadership, culture, and accountability while others report lack of support, unclear expectations, and a focus on metrics over leadership. Incentive structures discourage high-performing technicians from moving into management.
- **Strengths:**
  - Many GMs and MMs report that high performance and accountability are modeled and reinforced, indicating intent and awareness at senior levels.
  - Expectations are clear on paper. Most leaders, TSMs, and CSMs report a shared understanding of job expectations, quality definitions, and invoice accuracy standards.
  - Some sites use the performance management system and goal-setting effectively, localizing objectives and using performance conversations to influence behavior.
  - A majority of technicians report feeling comfortable raising ideas or concerns with their leads and managers, suggesting some baseline level of psychological safety.

  - CSMs generally see feedback on invoice accuracy and are helpful and timely, indicating pockets of strong performance management around that particular outcome.
  - Some leaders indicated they had taken performance management system training at headquarters, so leadership training already exists and is available.
- **Weaknesses:**
  - Clear perception gap between management and frontline staff. Leaders rate modeling and accountability high, but less than half of technicians feel high performance and accountability are actually modeled and reinforced.
  - Feedback is often infrequent, vague, or primarily critical. Many technicians report getting little actionable guidance on how to improve or progress.
  - Underperformance is tolerated without visible consequences, while strong performers are treated the same as those who underperform, lowering motivation and fairness.
  - Some employees describe expectations as unclear or shifting, leading to confusion over "what good looks like" day-to-day.
  - Pay and progression structures can disincentivize high performers from advancing; in some cases, moving into leadership means earning less than as a senior tech, which discourages high performers from taking on roles that require managing performance and culture.
  - Metrics are emphasized more than learning or outcomes. Targets are monitored, but employees experience feedback as "hit the numbers or be criticized," with limited focus on understanding causes, removing barriers, or recognizing high-quality work. This encourages surface-level compliance rather than genuine performance improvement.
- **Opportunities:**
  - Corporate HR teams and external talent management resources can provide structured, frontline leadership and performance management frameworks.
  - Increased focus on technician and manager retention across the aviation industry. LMNOP can treat leadership development as a way to attract and retain talent.
  - Compensation benchmarking and career-path work can realign pay and role design so leadership is desirable for high performers.
- **Threats:**
  - Competitive aviation organizations offering higher wages, advanced technology, and clearer career paths.
  - General aviation labor shortages increase the risk of losing experienced talent.
  - If leadership roles remain financially and culturally unattractive, it will be difficult to staff and retain strong leadership and TSMs, limiting the impact of other interventions.
- **Impact:** Technician performance, retention, and morale vary across service centers due to inconsistent leadership, training, and performance expectations. As a result, there are productivity issues, reduced communication, and increased operational strain and rework.

- **Levers:**
  - Define a simple, network-wide performance management standard (expectations, feedback cadence, recognition, and accountability) and embed it in performance management system and site routines.
  - Invest in frontline leadership development for TSMs and leads, with practical skills for coaching, setting local goals, and addressing underperformance
  - Review and adjust pay bands and career paths so moving into leadership is financially and professionally attractive for strong technicians
  - Rebalance performance conversations from “just the numbers” to metrics, causes, and behaviors, using metrics as a starting point for problem solving, rather than only for criticism.
  - Strengthen recognition practices that highlight quality, effective coaching, and thorough invoice review, not just volume or speed, and use cross-site comparisons to spread good practices from strong cultures to weaker ones.

Multicriteria Analysis

Criteria explained

1. **Cost –** *5 = Low cost, 1 = High cost.*

   How expensive would it be to develop and implement the intervention?

2. **Work Stoppage** – *5 = No disruption, 1 = Complete stoppage.*

   Would implementing the intervention cause any interruption to normal operations?

3. **Impact to Business (ROI)** – *5 = Very high ROI, 1 = Minimal or no ROI.*

   Will the intervention significantly improve NOP, productivity, or invoice accuracy?

4. **Speed of Implementation & Results** – *5 = Noticeable results within a few months, 1 = Little to no results in that timeframe.*

   How quickly can the intervention be implemented and start delivering measurable outcomes?

5. **Sustainability** – *5 = Highly sustainable, 1 = Difficult to maintain.*

   Is the intervention easy to maintain and keep up to date over time?

Table 8
Completed Multicriteria Analysis

Intervention	Criterion 1 rating: Cost	Criterion 2 rating: Work Stoppage	Criterion 3 rating: Speed of Implementation with Results	Criterion 4 rating: Sustainability	Criterion 5 rating: Impact to Performance Metric(s)	Criterion 6 rating: Number of Locations Affected	Avg rating
Weighted %	**20**	**25**	**10**	**5**	**20**	**20**	
Process Design (3.4)							
Standardize Operating Procedures	4	4	3	3	5	5	4.25
Process Management (3.3)							
Create a centralized KPI dashboard	4	4	2	3	1	5	3.35
Develop standardized escalation protocols	4	4	3	4	2	5	3.7
Knowledge & Training (3.4)							
Role Specific Onboarding Paths	4	4	3	4	5	5	4.3
Formal Mentorship Program	4	3	4	4	5	5	4.15
Bridge shifts	2	4	3	2	3	5	3.4
Rotating KSA Champions	4	5	3	4	5	5	4.55

Simulation based training	2	2	2	3	3	5	2.85
Peer learning shadowing	3	4	4	4	5	5	4.2
Frontline leadership development	4	3	3	3	5	5	4
Targeted Training	4	4	3	4	5	5	4.3
Information (3.8)							
Competency Model	4	4	3	4	4	5	4.1
Regular one-on-one meetings	4	3	4	3	5	5	4.1
Skills matrices	4	5	4	4	5	5	4.65
Mobile app with live work updates	2	4	2	4	2	5	3.2
Change management for system updates	4	5	5	4	4	5	4.5
Create and embed handoff templates	5	5	4	4	5	5	4.66
Motives (3.9)							
Career pathways leadership development program	4	5	4	4	5	5	4.65

Incentives (3.2)							
Recognition programs	3	5	5	4	5	5	4.55
Leadership compensation review	3	5	3	4	5	5	4.35
Resources (2.9)							
AI powered invoicing	2	4	2	4	2	5	3.2
Centralized information hub (RAG LLM or Knowledge Hub)	4	4	3	3	5	5	4.25
Assessment of digital tools	3	4	3	3	5	5	4.05
Create Job Aids	5	5	5	3	4	5	4.5
Single system of record	5	5	4	5	5	4	4.67
Update outdated tools	3	5	3	3	3	3	3.5
Wifi connectivity	4	3	4	4	5	3	3.75
Parts racks	5	5	4	5	2	2	3.7
Sheds	5	5	4	5	2	2	3.7
facility capacity and size	3	3	3	5	5	3	3.5

*Scale: 1 = very low, 2 = low, 3 = medium, 4 = high, and 5 = very high

B.2 Interview Information

Background

Problem statement: LMNOP Aviation is working to improve customer experience, efficiency, capacity, and profitability across its service centers. Currently, a majority of centers are underperforming in productivity and invoice accuracy. A key issue appears to be the lack of standardized routines and processes, which leads to wasted time, higher labor costs, inconsistent service, and billing errors. These inefficiencies not only impact profitability but also risk damaging their reputation and driving customers to competitors. We're here to ask questions and gain a better understanding of what's happening at your site. Your insights are essential to identifying opportunities for improvement and helping shape solutions that work across the network.
Main topic: What are the main factors that contribute to meeting or not meeting productivity and invoice accuracy goals?

Exemplary Locations

Exemplary Locations

Excels at productivity and Invoice Accuracy resulting in on-time maintenance, job quality (customer experience), NOP, and planning.

Questions
Excels at productivity and Invoice Accuracy resulting in on-time maintenance, job quality (customer experience),NOP, and planning.

- Can you walk me through your center's end-to-end service process,from aircraft arrival to departure?
- What is your process for planning and executing workscopes to adhere to schedule?
  - Before?
  - During?
  - After?
  - How do you assign work to ensure the schedules are met?
    - Do you use any tools for this?
- What systems or methods do you use to monitor and support technician productivity?
  - What systems or communication methods do you use to keep jobs on track?
  - How do you handle situations where the project is at risk of not meeting the deadlines?
  - What methods do you or your team use to communicate job status and notify leadership of potential delays or schedule changes?
- What are the top one or two things your team does that result in such high-quality work?

- What slows down your work or causes delays that prevent your team from working or releasing the A/C on time?
    - Are there recurring issues with equipment, processes, communication, etc.?
- What challenges do you face with scheduling, forecasting, and managing work?
    - How do you overcome them?
- What is your process for when a customer reports an issue or is dissatisfied with a service?
- What would help you be more effective and efficient in your role?
    - What about your team?
- What is one thing that you think your site does that helps ensure customer expectations are met?
- What is one area where you think your site could improve and why?
- What metrics does leadership target at your site?
    - Do you know why and how to meet their expectations for those metrics?
- What is one way you think your site could improve productivity?
    - Invoice accuracy?
- What role does leadership play in enabling your success?
- How do you onboard new technicians to maintain your standards?

Needs Improvement Locations

Questions

- Can you walk me through your center's end-to-end service process, from initial customer request to job close-out?
- What is your process for planning and executing workscopes to adhere to schedule?
    - Before?
    - During?
    - After?
    - Do you assign work to ensure the schedules are met?
        - Do you use any tools for this?
- What systems or methods do you use to monitor and support technician productivity?
    - What systems or communication methods do you use to keep jobs on track?
    - How do you handle situations where the project is at risk of not meeting the deadlines?
    - What methods do you or your team use to communicate job status and notify leadership of potential delays or schedule changes?
- What slows down your work or causes delays that prevent your team from working or releasing the A/C on time?
    - Are there recurring issues with equipment, processes, communication, etc.?

- What is your process for when a customer reports an issue or is dissatisfied with a service?
- What would help you be more effective and efficient in your role?
    - What about your team?
- What challenges do you face with scheduling, forecasting, and managing work?
    - How do you overcome them?
- What are the top one or two things your team does that results in high-quality work?
    - Do you do it every time?
    - What would help you do it every time?
- What is one area you think your site could improve and why?
- What metrics does leadership target at your site?
    - Do you know why and how to meet their expectations for those metrics?
- What factors do you think play a part in your site's ability to meet productivity or invoicing goals?
    - What is one way you think your site could improve productivity?
    - Invoice accuracy?
- What’s one process you feel is unclear or inconsistent?
- If you had one extra resource: time, staff, tool; what would make the biggest difference for you and your team?

B.3 Survey Questions

Should we put some kind of explanation here?

GM

Focus: Performance, metrics, customer experience

Instructions: Please answer the following questions based on your experience at your service center. Your honest feedback is crucial for helping us understand and improve our processes across the network.

Part 1: Performance

1. To what extent do you agree with the following statements?
(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

- I have a clear and shared understanding of job expectations for my team

- I am aware of how my daily work contributes to achieving my site's performance metrics and organizational goals.
- My site has the proper staff, equipment, and facilities to perform operations
- My employees receive the training they need to perform their roles effectively
- My site consistently meets our operational targets without sacrificing quality or safety

2. In a typical month, how often do the following occur at your site?
(Scale: 1 = Never, 2 = A couple times a month , 3 = Once a week, 4 = Multiple days a week, 5 = Every day)

- Waiting on parts or materials
- Unplanned or emergency jobs disrupting the schedule
- Insufficient technician availability
- Lack of necessary tools or equipment

3. What challenges do you face with scheduling, forecasting, and managing work?
    - How do you overcome them?

(Required) Open Ended

(Optional) Please explain any of your ratings above. What led you to choose those numbers? Specific examples are helpful. Please avoid names or confidential details.

Part 2: Motivation/Incentives

1. To what extent do you agree with the following statements?
(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

- My site recognizes and rewards those who meet performance goals
- The incentives provided by the organization motivate my performance
- High performance and accountability are modeled and reinforced by leadership
- Constructive feedback and recognition are part of my site's culture

(Optional) Please explain any of your ratings above. What led you to choose those numbers? Specific examples are helpful. Please avoid names or confidential details.

Part 3: Customer Experience

1. To what extent do you agree with the following statements?
(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

- My site uses customer feedback to improve our processes and service quality
- My site consistently delivers on our customer commitments
- My team understands how their work affects the customer experience

2. In your opinion, what is the single most important driver of high job quality and customer satisfaction?
(Required) Open Ended

3. What are the key factors that most influence your site's success?
(Required) Open Ended

(Optional) Please explain any of your ratings above. What led you to choose those numbers? Specific examples are helpful. Please avoid names or confidential details.

Part 4: Metrics

1. To what extent do you agree with the following statements?
(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

- I receive timely reports/data that help me identify areas of improvement
- My organization has a clear expectation regarding performance goals
- My site consistently meets or exceeds expectations for NOP
- My site consistently meets or exceeds expectations for productivity
- My site consistently meets or exceeds expectations for invoice accuracy
- Performance and customer experience metrics are discussed regularly with my team

(Optional) Please explain any of your ratings above. What led you to choose those numbers? Specific examples are helpful. Please avoid names or confidential details.

MM

Focus: Performance, metrics, technical operations

Instructions: Please answer the following questions based on your experience at your service center. Your honest feedback is crucial for helping us understand and improve our processes across the network.

Part 1: Performance

1. To what extent do you agree with the following statements?
(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

- I have a clear and shared understanding of job expectations for my team.
- I am aware of how my daily work contributes to achieving my site's performance metrics and organizational goals.
- My site has the proper staff, equipment, and facilities to perform operations
- My employees receive the training they need to perform their roles effectively
- My site consistently meets our operational targets without sacrificing quality or safety

2. In a typical month, how often do the following occur at your site?
(Scale: 1 = Never, 2 = A couple times a month , 3 = Once a week, 4 = Multiple days a week, 5 = Every day)

- Waiting on parts or materials
- Unplanned or emergency jobs disrupting the schedule
- Insufficient technician availability
- Lack of necessary tools or equipment

3. What challenges do you face with scheduling, forecasting, and managing work? How do you overcome them?
(Required) Open Ended

(Optional) Please explain any of your ratings above. What led you to choose those numbers? Specific examples are helpful. Please avoid names or confidential details.

Part 2 Motivations/Incentives

1. To what extent do you agree with the following statements?
(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

- My site recognizes and rewards those who meet performance goals
- The incentives provided by the organization motivate my performance
- High performance and accountability are modeled and reinforced by leadership
- Constructive feedback and recognition are part of my site's culture

- Performance reviews and recognition programs accurately reward employees

(Optional) Please explain any of your ratings above. What led you to choose those numbers? Specific examples are helpful. Please avoid names or confidential details.

Part 3: Metrics

1. To what extent do you agree with the following statements?
(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

- I receive timely reports/data that help me identify areas of improvement for my team
- My site has clear expectations regarding performance goals
- My site consistently meets or exceeds expectations for NOP
- My site consistently meets or exceeds expectations for productivity
- My site consistently meets or exceeds expectations for invoice accuracy
- Site leadership regularly discusses performance and customer experience metrics with their teams

(Optional) Please explain any of your ratings above. What led you to choose those numbers? Specific examples are helpful. Please avoid names or confidential details.

Part 3: Technical Operations

1. To what extent do you agree with the following statements?
(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

- My site's procedures are standardized and consistently followed across shifts and teams.
- Technical documentation (i.e. manuals, service bulletins, work instructions) are accurate, up-to-date, and easily accessible.
- We have effective processes in place for troubleshooting and resolving recurring technical issues.
- Cross-functional collaboration between maintenance, engineering, and quality teams supports efficient technical operations.
- We proactively identify and avoid risks related to maintenance.
- My team receives timely updates and training on new technologies and system changes.
- My team receives timely updates and training on new or changed processes and procedures.
- My site's technical operation directly contributes to minimizing aircraft downtime and improving turnaround time.

2. What change would have the greatest positive impact at your site?
(Required) Open Ended

(Optional) Please explain any of your ratings above. What led you to choose those numbers? Specific examples are helpful. Please avoid names or confidential details.

CSM

Focus: Invoice Accuracy, customer experience, and communication.

Instructions: Please answer the following questions based on your experience at your service center. Your honest feedback is crucial for helping us understand and improve our processes across the network.

Part 1: Invoice Accuracy Clarity and Communication

1. To what extent do you agree with the following statements?
(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

- Expectations for invoice accuracy are clearly defined and communicated
- Feedback from my leadership on invoice accuracy issues is helpful and timely
- I understand the components and standards that define a complete and accurate invoice
- Customers receive a timely explanation of billing details and corrections when needed
- Communication at my site follows a standard process that keeps me informed

2. What do customers most often ask or complain about in regards to billing at your location?
(Required) Open Ended

(Optional) Please explain any of your ratings above. What led you to choose those numbers? Specific examples are helpful. Please avoid names or confidential details.

Part 2: Resources

1. To what extent do you agree with the following statements?
(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

- The invoicing systems and processes support accurate billing and data matches (i.e. part component codes, coverage terms, standard labor hours, etc.)
- The invoicing systems and processes are standardized and easy to complete
- I have access to clear job aids to support accurate billing and data matches (i.e. step-by-step guide with component code definitions, standard plan hours, etc.)
- I have access to a checklist for reviewing invoices
- I have access to reliable and accurate data from my supporting teams (i.e. TSMs, technicians, and parts)
- My team is adequately staffed and we have time to review invoices for accuracy

- My workloads are reasonable

2. Does your site have a standardized process for reviewing invoices for accuracy? If so, what is the process?
(Required) Open Ended

(Optional) Please explain any of your ratings above. What led you to choose those numbers? Specific examples are helpful. Please avoid names or confidential details.

Part 3: Motivation/Incentives

1. To what extent do you agree with the following statements?
(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

- Leadership emphasizes billing and documentation accuracy when evaluating my performance.
- When invoice errors occur, leadership responds fairly and with appropriate accountability.
- I understand the impact of billing errors on customer experience and company finances.
- Performance reviews and recognition programs accurately reward employees
- I take pride in ensuring that every invoice is accurate
- I am aware of how my daily work contributes to achieving my site's performance metrics and organizational goals.
- High performance and accountability are modeled and reinforced by leadership
- Performance and customer experience metrics are discussed regularly with my team

(Optional) Please explain any of your ratings above. What led you to choose those numbers? Specific examples are helpful. Please avoid names or confidential details.

Part 4: Knowledge and Training

1. To what extent do you agree with the following statements?
(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

- I received the training I needed to perform my role effectively
- I understand the needs and expectations of my role
- I feel confident in performing the needs and expectations of my role.
- I receive timely updates and training on new technologies and system changes.
- I receive timely updates and training on new or changed processes and procedures.
- I am trained to identify and correct invoice errors
- The process for invoice approval and corrections is clearly documented and easy to follow.
- There are opportunities for me to develop my skills and advance

(Optional) Please explain any of your ratings above. What led you to choose those numbers? Specific examples are helpful. Please avoid names or confidential details.

Part 5: Customer Experience

1. To what extent do you agree with the following statements?
(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

- My site uses customer feedback to improve our processes and service quality
- My site consistently delivers on our customer commitments
- I understand how my work affects the customer experience directly

2. In your opinion, what is the single most important driver of high job quality and customer satisfaction?
(Required) Open Ended

2. In your opinion, what training or support would best help you or your team improve billing accuracy?
(Required) Open Ended

(Optional) Please explain any of your ratings above. What led you to choose those numbers? Specific examples are helpful. Please avoid names or confidential details.

TSM

Focus: Team performance,productivity, and technical operations.

Instructions: Please answer the following questions based on your experience at your service center. Your honest feedback is crucial for helping us understand and improve our processes across the network.

Part 1: Performance

1. To what extent do you agree with the following statements?
(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

- I have a clear and shared understanding of what defines a “high-quality” job
- I am aware of how my daily work contributes to achieving my site's performance metrics and organizational goals.
- My site has the proper staff, equipment, and facilities to perform operations
- My site consistently meets our operational targets without sacrificing quality or safety

2. In a typical month, how often do the following occur at your site?
(Scale: 1 = Never, 2 = A couple times a month , 3 = Once a week, 4 = Multiple days a week, 5 = Every day)

- Waiting on parts or materials
- Unplanned or emergency jobs disrupting the schedule
- Insufficient technician availability
- Lack of necessary tools or equipment

3. What change would have the greatest positive impact at your site?
(Required) Open Ended

(Optional) Please explain any of your ratings above. What led you to choose those numbers? Specific examples are helpful. Please avoid names or confidential details.

Part 2: Motivation/Incentives

1. To what extent do you agree with the following statements?
(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

- My efforts are recognized and appreciated by leadership
- Good performance is rewarded fairly
- I feel motivated to do my best work every day
- I take pride in the quality of the work I perform
- The work I do makes a meaningful contribution to the team's success
- High Performance and accountability are modeled and reinforced by leadership
- Performance reviews and recognition accurately reward employees

(Optional) Please explain any of your ratings above. What led you to choose those numbers? Specific examples are helpful. Please avoid names or confidential details.

Part 3: Productivity

1. To what extent do you agree with the following statements?
(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

- My team consistently achieves productivity goals
- I have the tools and resources to perform my job efficiently
- Processes at my service center are efficient
- I can complete my work without delays or interruptions
- My workloads are reasonable
- I received timely reports/data that help me identify areas of improvement for my team
- Performance and customer experience metrics are discussed regularly with my team

Please explain any of your ratings above. What led you to choose those numbers? Specific examples are helpful. Please avoid names or confidential details.

Part 4: Knowledge and Training

1. To what extent do you agree with the following statements?
(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

- I have received the training I need to perform my role effectively
- I know where to go for help or guidance when needed
- Ongoing training is available to accommodate changes in the system or processes
- There are opportunities for me to develop my skills and advance

Please explain any of your ratings above. What led you to choose those numbers? Specific examples are helpful. Please avoid names or confidential details.

Part 5: Technical Operations

1. To what extent do you agree with the following statements?
(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

- My site's procedures are standardized and consistently followed across shifts and teams.
- Technical documentation (i.e. manuals, service bulletins, work instructions) are accurate, up-to-date, and easily accessible.
- We have effective processes in place for troubleshooting and resolving recurring technical issues.
- Cross-functional collaboration between maintenance, engineering, and quality teams supports efficient technical operations.
- We proactively identify and avoid risks related to schedule adherence (i.e. part delays, troubleshooting delays, overclocking, etc.)
- My team receives timely updates and training on new technologies and system changes.
- My team receives timely updates and training on new or changed processes and procedures.
- My site’s technical operation directly contributes to minimizing aircraft downtime and improving turnaround time.
- Communication at my site follows a standard process that keeps me informed

(Optional) Please explain any of your ratings above. What led you to choose those numbers? Specific examples are helpful. Please avoid names or confidential details.

Technicians

Focus: Productivity, process, and communication

Instructions: Please answer the following questions based on your experience at your service center. Your honest feedback is crucial for helping us understand and improve our processes across the network.

Part 1: Productivity

1. To what extent do you agree with the following statements?
(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

- My site's performance expectations are clearly communicated to me
- I consistently receive feedback about my ability to achieve productivity goals
- I have the tools and resources to perform my job efficiently
- Processes at my service center are efficient
- My workloads are reasonable
- I can complete my work without unnecessary delays or interruptions

2. When you're working on a job, how often do you have what you need to do your best work?
(Scale: 1 = Never, 2 = Rarely, 3 = Sometimes, 4 = Often, 5 = Always)

- Access to up-to-date and accurate technical manuals
- The right tools for every job, in good working condition
- The correct parts, available when needed
- Support from a lead technician, TSM, or expert when questions arise

Please explain any of your ratings above. What led you to choose those numbers? Specific examples are helpful. Please avoid names or confidential details.

Part 2: Motivations/Incentives

1. To what extent do you agree with the following statements?
(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

- My efforts are recognized and appreciated by leadership
- Good performance is rewarded fairly
- I feel motivated to do my best work every day
- I take pride in the quality of the work I perform
- The work I do makes a meaningful contribution to the team's success
- I am aware of how my daily work contributes to achieving my site's performance metrics and organizational goals.
- High performance and accountability are modeled and reinforced by leadership
- Performance and customer experience metrics are regularly discussed with my team
- Performance reviews and recognition programs accurately reward employees

2. What change would have the greatest positive impact at your site?

Please explain any of your ratings above. What led you to choose those numbers? Specific examples are helpful. Please avoid names or confidential details.

Part 3: Communication

1. To what extent do you agree with the following statements?
(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

- My supervisor communicates performance expectations clearly
- I receive feedback that helps me improve my performance
- I feel comfortable sharing ideas or concerns with my supervisor
- Communication at my site follows a standard process that keeps me informed

(Optional) Please explain any of your ratings above. What led you to choose those numbers? Specific examples are helpful. Please avoid names or confidential details.

Part 4: Processes

1. To what extent do you agree with the following statements?
(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

- Work processes at my service center are clearly defined and easy to follow
- Equipment and tools are maintained and in good working condition
- When process-related changes happen, they are explained clearly and in a timely manner
- There is consistency in how jobs are completed across shifts or teams
- I can easily complete tasks left by the previous shift

2. Which of the following factors most often affect your ability to complete a job with accuracy and efficiency?
(Select all that apply)

- Work instructions that need more clarity or detail
- Tight time expectations or competing priorities
- Waiting for the right parts or tools
- Being asked to assist with urgent or unexpected tasks
- Other (please specify): ________________________

3. In a typical week, how often do the following occur in your work area?
(Scale: 1 = Never, 2 = A couple times a month , 3 = Once a week, 4 = Multiple days a week, 5 = Every day)

- Waiting on parts

- Being redirected to an unplanned or urgent job
- Searching for the right tools or equipment
- Needing clarification on work instructions
- Waiting for an inspection or sign-off

(Optional) Please explain any of your ratings above. What led you to choose those numbers? Specific examples are helpful. Please avoid names or confidential details.

Part 5: Knowledge and Training

1. To what extent do you agree with the following statements?
(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

- I have received the training I need to perform my role effectively
- I know where to go for help or guidance when needed
- Ongoing training is available to accommodate changes in the system or processes
- There are opportunities for me to develop my skills and advance

(Optional) Please explain any of your ratings above. What led you to choose those numbers? Specific examples are helpful. Please avoid names or confidential details.

Appendix C: Ethical Treatment

This needs assessment will be conducted in accordance with the ISPI Code of Ethics (Add Value, Validated Practice, Collaboration, Continuous Improvement, Integrity) ISPI Performance Standards, and the Belmont Principles (respect for persons, beneficence, and justice).

Participant Recruitment and Informed Consent

- Participation in surveys, interviews, and other data collection methods will always be voluntary, and non-coercive recruitment tactics will always be used.
- There will never be a penalty for non-participation.
- All data collection efforts will begin with an informed consent document (see Appendix), stating the following:
    - The purpose of the data collection
    - Who is collecting the data
    - How the data will be used
    - How the data collection method will proceed
    - How the participants' confidentiality will be maintained, including data handling
    - Any risks or benefits to the participant
    - Time commitment
    - Their participation is voluntary
    - Who to contact with any concerns or questions
    - Right to refuse or withdraw at any time

Data Minimization & Scope

- Only the minimum amount of data that is necessary to understand and define the gap, and make decisions on interventions will be collected.
    - For example, names of participants will not be collected, and data such as service center sites are anonymized.

Collection Protocols

- **Surveys** will be anonymous by default. Open text fields will only be used if necessary to avoid inadvertently collecting PII.

- **Interviews/focus groups** will be conducted using a guide (either structured or semi-structured), and no supervisors will be present. We will reiterate confidentiality at the start. Recording can be refused or stopped at any time. No names or other identifying information will be recorded. The roles of each interviewer will be clearly identified (i.e., whether they are internal or external to LMNOP Aviation, as well as their role at the organization they're representing). Interviewees have the right to request that any interviewers be removed from the process, and that right will be granted.

- **System/Document data** will be anonymized to everyone except the team lead.

Privacy, Confidentiality, and De-identification

- We will use unique anonymized identifiers in place of PII, or identifying information about sites.
  - For example: Location A, Location B
- Quotes containing identifying details will be paraphrased. Unique job titles will be masked.

Data Security & Access Control

- Only aggregated data will be shared with the client.
- Data will be retained until the final report is delivered to the client, then it will be destroyed.
- Consent forms will be retained for three years.

Use of Data Collection and Analysis Tools

- We will disclose all tools used (survey platforms, software, recording equipment).
- AI tools
  - We may use AI for brainstorming and general ideation.
  - We will not enter sensitive data (SD) into any AI system. SD includes any information that could harm a person or client if disclosed or misused, including metrics, processes, identifiable information, financials, customer or employee data, proprietary methods, strategy, and any material considered intellectual property.

Equity, Bias, and Accessibility

- We will ensure that data collection tools, such as surveys, are accessible.
- We will provide accommodations to interviewees who need them to ensure equitable access and participation.
- If sampling takes place, particular efforts will be made to include a diverse range of perspectives.
- We acknowledge that one member of the project team has prior aviation experience, which could influence their perspective on the organization or its sites. To mitigate this, the remainder of the team will act as a control and maintain objectivity. As external investigators, we do not hold any preference toward specific sites.

Analysis Integrity and Quality

- Address missing data transparently by maintaining a log or adding it to the limitations section of the analysis plan.

Transparency in Reporting

- We will report our methods and limitations, including our reasoning for not including specific data.
- We will outline how we created our recommendations based on evidence and ISPI's performance standards.

C.1 Interview Script

Background

Problem statement: LMNOP Aviation is working to improve customer experience, efficiency, capacity, and profitability across its service centers. Currently, a majority of centers are underperforming in productivity and invoice accuracy. A key issue appears to be the lack of standardized routines and processes, which leads to wasted time, higher labor costs, inconsistent service, and billing errors. These inefficiencies not only impact profitability but also risk damaging their reputation and driving customers to competitors. We're here to ask questions and gain a better understanding of what's happening at your site. Your insights are essential to identifying opportunities for improvement and helping shape solutions that work across the network.

Main topic: *What are the main factors that contribute to meeting or not meeting productivity and invoice accuracy goals?*

Opening Script

Hello, nice to meet you, my name is__________. My team and I are currently pursuing M.S. in Organizational Performance and Workplace Learning at Boise State University and are currently enrolled in a Needs Assessment course. We would first like to start by thanking you for agreeing to take time out of your day to participate in this interview. We will be using the information gathered from this interview for our needs assessment class project. We invited you to do this interview because we think you will provide a unique perspective with your given experience. This interview will take about 30 minutes and will cover approximately 11 questions with some deeper dives as needed. We want to reiterate what was explained in the informed consent document that you signed; this interview can be stopped or postponed at any time, we want you to feel comfortable and willing to share. We will be using this information for our project but your name/identity will be protected and remain confidential. Now we must ask and confirm again that you are aware that this video will be recorded for transcription purposes only. Now that we have covered all of that, are you ready to begin the interview?

Interview Questions

Begin recording meeting

C.2 Informed Consent

Informed Consent Form

STUDY TITLE: Needs Assessment Analysis LMNOP Aviation

INVESTIGATOR: Sarah Chapin, Ana Jackson, Michelle Sinn, and Patrick Smith (Evaluation Team from the OPWL Graduation Program).

SPONSOR: Blake Harrison, Talent and Development Manager, LMNOP Aviation

APPROVAL: This needs assessment project has been approved by LMNOP Aviation's domestic service center leadership

PURPOSE: Identify and address performance inconsistencies in LMNOP Aviation's domestic service centers by evaluating roles, processes, and outcomes to improve the organization's productivity and invoice accuracy.

PROCEDURE: Team Aviation Practitioners will contact you via email to schedule an interview date and time. The interview will be conducted via Zoom and will last for about 30 minutes. With your permission, we will video-record the interview for transcription only, and then we will destroy the file after the transcript is complete.

VOLUNTARY PARTICIPATION: Your participation in this needs assessment project is voluntary. Participation is voluntary. You may decline or stop at any time without penalty. Your choice will not affect your job, performance review, pay, benefits, promotion opportunities, work assignments, or employment status.

CONFIDENTIALITY: The needs assessment team will keep any potentially identifiable information confidential. Your individual responses will not be shared with your supervisor or anyone in your management chain. Only a summary of data will be presented in a report, and your name will not be used in any written reports or publications that result from this needs assessment.

RISKS: There are minimal risks associated with this interview. If you do not feel comfortable answering any of the questions during the interview, you can skip the question or stop participating in the interview at any time.

BENEFITS: By submitting your input, you are sharing helpful information that will provide the data needed to identify and define solutions to improve overall performance in the domestic service center network.

QUESTIONS: If you have any questions or concerns about participation in this needs assessment project, please contact the needs assessment project team sarahchapin@u.boisestate.edu, anajackson@u.boisestate.edu, patricksmith@u.boisestate.edu, michellesinn@u.boisestate.edu. The professor and advisor for this project, Don Winiecki, can be reached at dwiniecki@u.boisestate.edu.

By typing my full name below, I confirm that I am 18 or older, have read and understood this form, and agree to participate voluntarily. I understand I may withdraw at any time without penalty.

- Typed Name: ___________________________
- Date: ____ / ____ / ______

C.3 Email Survey Script

To: All groups
Subject: Service Center Feedback Survey

Hello,

My name is ___________, and I am currently pursuing a Master's degree in Organizational Performance and Workplace Learning at Boise State University. As part of my coursework, I am leading a team of graduate students on a project to identify performance needs across the service network.

We are reaching out to gather feedback from various roles and locations to better understand how work is accomplished across sites. Your input is essential for identifying opportunities to improve processes and overall performance.

There are minimal risks associated with this survey, and all responses will be anonymized and kept confidential. By participating, you will help provide valuable data that will inform recommendations for improving productivity and accuracy across the network.

The survey should take about **5 minutes** to complete and is due by **November 17th at 3:00 PM**. If you have any questions, please feel free to contact me via email.

Thank you for your time and participation!

Appendix D: Needs Assessment Planning Table

Problem or Opportunity

Dates	**Start: 9/18/2025** **End: 11/7/2025**
Key Questions	• What is the main driver for this request at Aviation Company LMNOP? • What is the desired performance outcome or outcomes? • How will we know when we have solved this problem or set of problems? • Is this a performance problem? • What are the specific performance metrics showing a problem? (e.g., productivity, invoice accuracy) • What was the initial proposed solution, and why did it fail?
Framework/Model **(listed in order of use)**	Harless Front End Analysis
Data Sources	Client: Blake Harrison Executives/leaders: Specify: Cameron Steele Target population: Specify: SMEs / stars Specify: GMs Upstream stakeholders Specify: Downstream stakeholders Specify: customers

Data Collection Methods	Extant Data/Documents Specify: Invoice accuracy and productivity (driving metrics that are tied to our needs) From organization Specify: From the literature review Specify: Other case studies & best practices review Specify: Observation Specify: Interviews Individual Group Unstructured: Semi-structured Structured Specify: Survey Specify: Other:
Analysis	• **Need 1: Improve productivity** ○ Performance outcome: TSMs accurately plan hours worked, TSMs accurately schedule hours for revision, maximize utilization, TSMs hold techs accountable for unexcused absences or excessive idle time • **Need 2: Improve invoice accuracy** ○ Performance outcome: Techs create accurate invoices on first pass by performing at or below scheduled standard plan hours, CSMs catch errors before invoice is issued to customer

Notes	**Current State:** - Less than half the service centers meet threshold for productivity - About half the service centers meet threshold for invoice accuracy **Desired State:** 91% service centers meet threshold for productivity 91% service centers meet threshold for invoice accuracy

Environmental Scan

Dates	**Start: 10/8/2025** **End: 11/14/2025**
Key Questions	- What are the attrition rates for the employee pool for each service center? - Are there any external forces impacting the performance problem? (for instance, shipping times, sourcing parts or other resources, etc)? - What do customers expect and desire when they bring their jet in for maintenance?
Framework/Model (listed in order of use)	- Rummler and Brache's Nine-Variable Model (CSM/TSM/MM middle row) - Rummler and Brache's Nine-Variable Model (GMs top row) - Chevalier's BEM for Techs
Data Sources	**Client:** Blake Harrison **Executives/leaders:** Specify: Locational General Managers and Maintenance Managers **Target population:** Specify:

	SMEs / stars Specify: TSM/CSM/Techs **Upstream stakeholders** Specify: **Downstream stakeholders** Specify:
Data Collection Methods	Extant Data/Documents Specify: Hiring/recruiting information From organization Specify: From literature review Specify: Other case studies & best practices review Specify: Observation Specify: Interviews Individual Group Unstructured: Semi-structured Structured Specify: Survey Specify: Convergent parallel design/mixed methods participants: GMs, MMs, TSMs,CSMs, and Techs Other:
Analysis	• There are some resource constraints on parts mostly due to global supply chain shortages • There are heavier attrition rates at underperforming sites. • Consistent Wifi and system outages at some sites.

Notes	• Originally we would have liked to have considered competitor data, but industry benchmarking is either unavailable or does not exist.

Organizational Analysis

Dates	**Start: 9/18/2025** **End: 11/14/2025**
Key Questions	• How is Aviation Company LMNOP organized, and does this structure relate to or impact the performance issue? • How does LMNOP's mission, vision, and values influence decision-making, project management, and performance priorities? • Are there any stated organizational goals related to operational excellence or process standardization? • How do goals from upper management flow down to the service centers? Is there a disconnect between corporate goals and service center reality? • What environmental and individual factors influence performance at the organizational and process levels at LMNOP Aviation? • What incentivizes the team members at each level, and how do those incentives reinforce or undermine performance expectations? • What motivates the team members at each level, and how is that motivation supported or hindered by current systems and structures at LMNOP Aviation? • What factors are within the organization's control, and what can be used to influence them? • What processes currently exist at LMNOP Aviation, and how effectively do they support desired performance outcomes?

	- How are key work processes designed? Are they efficient and standardized, or do they vary by location? Is there clear process ownership? - What information and resources are provided to performers to support desired accomplishments? - Is there an onboarding or mentoring program for managers? Does it include customer service skills? - Is there a set of tools or other supports that would help them learn and apply skills related to customer service? - What are the specific practices, workflows, or tools used by exemplary centers that are missing from non-exemplary centers?
Framework/Model (listed in order of use)	- Rummler and Brache's Nine-Variable Model (GM/CSMs/TSMss top row) - Chevalier's BEM for Techs
Data Sources	**Client:** **Executives/leaders:** Specify: TSMs/CSMs at two underperforming sites and two high-performing sites **Target population:** Specify: Technicians **SMEs / stars** Specify: **Upstream stakeholders** Specify: **Downstream stakeholders** Specify:
Data Collection Methods	Extant Data/Documents Specify: SOPs (if they exist), training or performance support materials regarding customer service (if it exists)

	From organization Specify: From literature review Specify: Other case studies & best practices review Specify: Observation Specify: Interviews Individual Group Unstructured: Semi-structured Structured Specify: Survey Specify: Convergent parallel design/mixed methods participants: GMs, MMs, TSMs,CSMs, and Techs Other:
Analysis	• There are no standard routines or processes across service centers. • Fragmented or undeveloped processes at underperforming service centers. • High workloads; diverse clientele • Lack of communication on performance • Management (TSMs & CSMs) are not holding teams accountable.
Notes	• Aviation Company LMNOP has a substantial number of supervisors managing an even larger technician workforce of various skill levels. • Supervisors and technicians have extensive experience in their roles.

Gap Analysis

Dates	**Start: 10/8/2025** **End: 11/2/2025**
Key Questions	• What is the current standard performance method and goal for managing projects at LMNOP Aviation? • What does ideal performance look like, and how is it defined? (exemplary vs. non-exemplary) • Where are the biggest disconnects between goals, design, and management at each level (organization, process, job)? • What forces are driving towards improved change and what forces are resisting it? • What factors are within our control, and which require external support or systemic change?
Framework/Model (listed in order of use)	• Rummler and Brache's Nine-Variable Model (CSM/TSM middle row) • Rummler and Brache's Nine-Variable Model (GMs top row) • Chevalier's BEM for Techs
Data Sources	**Client:** **Executives/leaders:** Specify: Cameron Steele, GMs, and MMs **Target population:** Specify: Technicians,TSMs, and CSMs **SMEs / stars** Specify: **Upstream stakeholders** Specify: **Downstream stakeholders** Specify:

Data Collection Methods	Extant Data/Documents Specify: Performance metrics pertaining to customer service, company website, internal K-Base, SOPs (if they exist), Training or performance support materials regarding customer service (if it exists) From organization Specify: From literature review Specify: Other case studies & best practices review Specify: Observation Specify: Interviews Individual Group Unstructured Semi-structured Structured Specify: Survey Specify: Convergent parallel design/mixed methods participants: GMs, MMs, TSMs,CSMs, and Techs Other:
Analysis	• Analyzed extant data for all service centers on productivity and invoice accuracy • Identified performance gaps: less that half centers meet productivity goals, only about half of centers meet invoice accuracy threshold. • Calculated gap between current state and desired state: a majority of service centers which are not meeting both thresholds. • Used data to determine 2 exemplary locations and 2 underperforming locations

	• Documented gap implications for strategic objectives: profitability (revenue leakage, increased labor costs), operational efficiency (unpredictable timelines, written off work), customer experience (delays, billing errors), capacity (constrained growth) • Established desired state goal: 91% centers meeting both productivity and invoice accuracy thresholds
Notes	• High number of sites not meeting both goals indicated root causes are systemic and not location-specific.

Cause Analysis

Dates	**Start: 10/8/2025** **End: 11/17/2025**
Key Questions	• What are the root causes for the gaps in invoice accuracy and productivity the organization is facing? • What evidence supports the existence of these gaps? • Is this truly a performance problem or lack of resources, incentives, or clarity? • Do technicians have clear expectations, necessary resources, and appropriate incentives? • Is the feedback time-sensitive? • Do performers have the necessary skills and knowledge to meet standards? • Do they know where to go for accurate and accessible information?
Framework/Model (listed in order of use)	• BEM • Rummler & Brache's Nine-Variable Model • Five Whys

Data Sources	**Client:** **Executives/leaders:** Specify: **Target population:** Specify: Technicians, TSMs, and CSMs **SMEs / stars** Specify: **Upstream stakeholders** Specify: **Downstream stakeholders** Specify:
Data Collection Methods	Extant Data/Documents Specify: From organization Specify: Metrics From literature review Specify: Other case studies & best practices review Specify: Observation Specify: Interviews Individual Group Unstructured: Semi-structured Structured Specify: Survey Specify: Convergent parallel design/mixed methods

	Other:
Analysis	Chevalier's BEM ● Lack of Information- No standardized processes for project planning, scheduling, or work allocation across network; expectations vary by location and manager ● Resource gaps - Inadequate/outdated tools (slow Wi-Fi, broken diagnostic computers, insufficient parts inventory); facility constraints (undersized hangars); some systems reported as slow, unreliable, difficult to use ● Knowledge/Skills gaps- Workforce transition from experienced veterans to less-experienced technicians without formal knowledge transfer mechanisms causes new hires to learn through "trial and error" rather than structured onboarding Rummler & Brache's Nine-Variable Model ● No standardized invoice review procedures, which causes CSMs to develop own workarounds rather than following documented processes ● CSM role expectations are unclear, with workload rated as unreasonable by 55% of CSMs. ● CSMs without aviation technical background struggle with technical reason codes and warranty coverage determinations
Notes	● Data suggests both high and low performers face similar systemic issues ● Exemplary locations have developed local workarounds while underforming locations haven't adapted

Intervention Selection/Feasibility Analysis

Dates	**Start: 11/12/2025** **End: 12/6/2025**
Key Questions	● What intervention or set of interventions offers the greatest value? ● How does each potential intervention align with the organization's strategic goals? ● What are the potential risks or barriers to implementing or not implementing each intervention?

	• Which interventions reinforce the drivers of performance or reduce the impact of barriers? • Which interventions would address any lack of resources, incentives, or clarity? • If lack of process is a contributing factor, which processes should be standardized?
Framework/Model (listed in order of use)	• SWOT • Multicriteria Analysis
Data Sources	**Client:** **Executives/leaders:** Specify: Cameron Steele, Blake Harrison, and GMs **Target population:** Specify: **SMEs / stars** Specify: **Upstream stakeholders** Specify: **Downstream stakeholders** Specify:
Data Collection Methods	Extant Data/Documents Specify: Primary metrics From organization Specify: From literature review Specify: Other case studies & best practices review Specify: Observation Specify: Interviews

	Individual Group Unstructured: GMs Semi-structured Structured Specify: Survey Specify: Other:
Analysis	• Organized interventions into three priority levels based on impact and organizational reach • Primary Interventions: Standardize core processes (SOPs for project planning, invoicing, work allocation), improve tools and systems (centralize documentation, assess to computer, improve reliability), formalize knowledge transfer (mentoring, role-specific onboarding, certification pathways) • Second Level Interventions: Strengthen performance management (train managers on feedback, establish recognition programs), develop career pathways (adjust compensation, leadership development, clear advancement criteria), enable peer learning (site champions, role shadowing, innovation recognition) • Third Level Interventions: Address site-specific infrastructure needs (Wi-Fi connectivity, parts racks, sheds, facility capacity at specific locations) • Selection Rationale: Multi-criteria analysis using weighted criteria (Impact, Feasibility, Readiness, Alignment, Scalability), confirmed primary interventions produce greatest network-wide improvement
Notes	• Met with core team of GMs to determine criteria • Primary recommendations address key root causes affecting both metrics across all locations • Second Level Recommendations address network-wide causes that are less impactful than primary recommendations • Third Level Recommendations are targeted interventions for individual sites